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ABSTRACT

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Ultrasound elastography of choroidal tumors: Initial experience

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Purpose: To report our initial experience with ultrasound elastography as a diagnostic tool in the differential diagnosis of choroidal tumors.
Methods: Ten eyes with various choroidal lesions were examined using the Clarius L20HD scanner in strain elastography mode. We included patients with choroidal melanoma, choroidal nevus, and choroidal detachment (solutio choroideae). Elasticity measurements of the lesions were compared with the surrounding normal choroidal tissue. Strain elastography was performed by applying gentle pressure with the ultrasound probe and analyzing the relative deformation of different tissue types in false colors.
Results: Choroidal melanomas demonstrated markedly decreased elasticity compared to the surrounding choroidal tissue, appearing stiffer on elastographic imaging. In contrast, choroidal nevi and choroidal detachments showed elasticity values comparable to the surrounding normal choroidal tissue, indicating similar mechanical properties.
Conclusions: Our initial experience suggests that ultrasound elastography may be helpful as an additional diagnostic tool in the differential diagnosis of choroidal tumors. The technique appeared particularly useful in distinguishing choroidal melanomas from benign lesions such as nevi and from non-neoplastic conditions such as choroidal detachment. The decreased elasticity of melanomas likely reflects their distinct histopathological composition and cellular density. Further studies with larger patient cohorts are needed to validate these preliminary findings and establish standardized elastographic criteria for choroidal lesions.

Shared decision making about choroidal melanoma treatment: Non-neutral framing of medical information

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Purpose: A subset of choroidal melanoma can be treated using either enucleation or proton beam therapy (PBT), which offer similar oncological outcomes. The most appropriate treatment depends on a patient's preference. To allow patients to determine their preference as genuinely as possible, it is recommended to describe options as neutrally as possible. This study assesses to what extent ocular oncologists use non-neutral framing.
Methods: Consultations of ocular oncologists with patients newly-diagnosed with choroidal melanoma at Leiden University Medical Center (LUMC) were audio recorded and transcribed verbatim. Transcripts were coded on explicit and implicit non-neutral framing behaviors. Explicit non-neutral framing was defined as: the ocular oncologist explicitly mentions a preferred option at least once without relating that preference to a preference of the patient. Implicit non-neutral framing was defined as: the ocular oncologist describes an option favorably or unfavorably, without providing a medically substantive clarification alongside.
Results: 110 patients provided consent for the audio recordings. Non-neutral framing was found in 84% (n=92/110) of consultations. We found explicit behavior in 38% (42/110) and implicit behavior in 76% (84/110, median = 1, range, 0–4) of consultations. The most frequent implicit behavior was presenting treatment options by emphasizing, in a positive or negative manner, the first or last treatment option. Non-neutral framing towards PBT was observed in 59% (54/92) of consultations.
Conclusion(s): This study shows that in most consultations some non-neutral framing was present, with a tendency to frame towards PBT. Ocular oncologists should be aware that the way they describe options may influence patients' treatment preference.

Outcomes of cataract surgery in patients with irradiated uveal melanoma treated with ruthenium-106 brachytherapy or proton beam therapy

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Purpose: To evaluate the visual, refractive and surgical outcomes of cataract surgery in patients with uveal melanoma (UM) previously treated with ruthenium-106 (Ru-106) brachytherapy or proton beam therapy (PBT).
Setting: Department of Ophthalmology, Leiden University Medical Center, Leiden, the Netherlands.
Design: Retrospective cohort study.
Methods: Patients with irradiated UM who underwent cataract surgery between 2015 and 2025 were included. Primary outcome measures were corrected distance visual acuity (CDVA) and refractive accuracy. Secondary outcomes included intraoperative and postoperative complication rates and additional secondary interventions.
Results: A total of 108 eyes were included (n = 72 Ru-106; n = 36 PBT). Mean CDVA improved to 0.48 ± 0.77 logMAR at 4 weeks (p < 0.001), with sustained 1-year gain following Ru-106, while CDVA declined in PBT-treated eyes due to radiation-induced complications. Eighty-three percent of eyes were within ±1.00 diopter (D) and 58.0% within ±0.50 D of target refraction. Intraoperative complications occurred in 12.0% of eyes, primarily following PBT. Persistent intraocular inflammation was a common postoperative complication, but decreased over time, while cystoid macular edema (CME) remained frequently observed. The probability of cataract surgery following PBT was significantly higher compared to Ru-106 (Log-Rank, p < 0.0001).
Conclusions: Cataract surgery significantly improves VA and offers favorable refractive accuracy in UM patients following radiation therapy. However, outcomes vary by treatment modality, with PBT-treated eyes demonstrating shorter time interval to cataract surgery, higher complication rates and less sustained long-term visual improvement.

Three years of ocular proton therapy in the Netherlands, clinical results

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Purpose: Uveal melanoma (UM) is a rare cancer with a rising incidence of 143 in 1990 to 226 in 2023 in the Netherlands. Most patients are treated locally with brachytherapy (±102 /year). Patients were referred for proton therapy (PT) to Switzerland in the past, but proton therapy is available in The Netherlands from Jan 2020 onwards. The purpose of this study is to assess the oncological and ophthalmological outcomes of the current cohort and compare with the published historic cohort.
Patients and Methods: All consecutive patients with PT for UM from Jan 2020 till December 31st 2023 (n = 176) and the Historic Cohort (n = 101) were treated with 4 × 15 CGE and were included. Time to event analyses were conducted according to the Kaplan-Meier method.
Results: The median age in the current cohort was 65 years (Historic: 59 p = 0.001). Tumor stage differed between the cohorts T1 current 10% (historic 3%), T2 21% (26%), T3 55% (36%), T4 14% (36%) (p = 0.004). Tumors regressed in 2-3 years to a remnant of around 3mm. At three years, OS was 78.9% (86%), DM 26.6% (17%), LF 3.5% (6%) and Enucleation Rate 16.8% (19%), the latter mostly because of pain and/or retinal detachment. In both cohorts visual acuity dropped substantially after treatment (median −0.50).At 3 years, in both cohorts around a quarter of the patients retained a vision of ≥0.3. Retinal Detachment was the most common toxicity in the current cohort with 52.3%.
Conclusion: The establishment of the HollandPTC improves access to proton therapy for Dutch uveal melanoma patients and results in additional sparing of around 40 eyes per year. Tumor regression after treatment happened gradually from 3 months to 2–3 years, often leaving a rest behind of 3–5 mm. Tumor control, eye preservation rates and side effects are similar to the published historic cohort treated abroad.

Local failure after ruthenium-106 plaque brachytherapy for uveal melanoma is correlated to institutes' annual treatment volume

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Introduction: Published local failure risks at five years after ruthenium-106 brachytherapy for uveal melanoma (UM) vary in the literature from 1%–42%. We noticed that centers that treat more patients per year reported lower local failure rates and therefore investigated the relation between a treating centers annual treatment volume (ATV) and the risk of local failure at five years after ruthenium-106 brachytherapy for UM.
Materials and methods: Systematic review of the literature and meta-analysis using a random effects model of all studies with at least 10 UM patients treated with ruthenium-106 brachytherapy, of which at least one treated after the year 2000, with a median follow-up of at least 3 years, that reported the local failure risk, comparing local failure between centers with a low (<30) vs high (≥30) ATV.
Results: Of 231 screened studies, we included 38 in full-text screening and 18 studies describing 4186 patients in the analyses. 6 studies were from centers that treated ≥30 patients per year (high ATV group), while 12 centers treated <30 patients per year (low ATV group). There was a wide variety in indications and treatment regimens. The pooled estimate 5-year local failure risk was 9.3% for the full cohort and was 4.1% in centers with a high (≥30) vs 15.2% in those with a low (<30) ATV ($p<0.001$).
Conclusion: In centers with a high annual treatment volume (≥30 annually), local failure risk at five years after ruthenium-106 brachytherapy for uveal melanoma is significantly lower (4%) than in centers with a lower annual treatment volume (15%).

Clipless ocular proton therapy

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Purpose: Proton beam therapy (PBT) is, alongside brachytherapy, a main eye-preserving treatment for uveal melanoma (UM). Currently, PBT requires surgical placement of scleral clips for tumor localization and patient positioning. This study introduces a novel clipless workflow for ocular PBT.
Methods: Patients receive an MRI and CT scan in an in-house developed immobilization setup with integrated gaze guidance. The MR-images are used for target definition and fused with the CT-images for Digital Reconstructed Radiographs (DRR), enabling bone-match-based positioning at the HollandPTC Eye Treatment Room. The accuracy of this positioning approach was evaluated using retrospective data of 60 UM patients. Additionally, a prospective end-to-end evaluation was performed using a head phantom, developed to simulate the clinical workflow, with the same clips placed on a 3D-printed eye.
Results: The immobilization setup showed minimal deformation between (supine) imaging and (sitting) treatment position (<0.5 mm). The retrospective evaluation of the bone-match positioning method showed negligible systematic errors and random positioning deviations of 0.4 mm with respect to the clips. The end-to-end accuracy assessment confirmed a precision of 0.4 mm in all directions.
Conclusions: The proposed clipless workflow is feasible and is currently being prospectively validated in the COPTER trial.

Worldwide incidence of ocular melanoma and correlation with pigmentation-related risk factors

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Purpose: To update the global incidence of ocular melanoma (OM), including conjunctival and uveal melanoma (UM). Light iris color and single-nucleotide polymorphisms (SNPs) have been described as UM-risk factors. Additionally, two of these are also associated with iris color (rs129138329 in HERC2 and rs12203592 in IRF4). We compiled worldwide OM incidence data and examined correlations with iris color and SNPs associated with iris color and/or UM-risk.
Methods: OM cases, defined by International Classification of Diseases Oncology codes C69 (eye), 8720/3–8790/3 (malignant melanoma), and 8000–8005 (malignant neoplasm), diagnosed between 1988 and 2012, were extracted from Cancer Incidence in Five Continents. Incidence rates were age-standardized, and temporal patterns were evaluated using joinpoint regression and age-period-cohort models. Country-level frequencies of iris colors, SNPs associated with iris color and/or UM-risk, were obtained from published sources.
Results: Incidence rates were ≥8.0 per million person-years in Northern Europe, Western Europe and Oceania; 2.0–7.9 in North America, Eastern Europe, and Southern

Europe; and <2.0 in South America, Asia, and Africa. OM incidence correlated strongly with latitude (r=0.77, p≤0.001), forming a north-to-south decreasing gradient across Europe. SNP rs12913832 showed significant correlations with OM incidence (r=0.83, p≤0.001), blue (r=0.56, p≤0.05) and green iris color (r=0.51, p≤0.05), and inversely with brown iris color (r=−0.64, p≤0.01). Incidence trends were stable in most countries (N=28/35).
Conclusions: OM incidence was highest in populations of European ancestry and lowest in Asian and African populations. Geographical correlation between OM incidence, iris color, and rs12913832 in HERC2 support the role of pigmentation-related risk factors in OM development.

Integrated clinical and in silico assessment of Von Hippel Lindau variant impact in retinal hemangioblastomas

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Introduction: This multicenter study evaluated associations between *VHL* gene variants and clinical outcomes in patients with retinal hemangioblastomas (RHs), focusing on tumor regression and treatment complications.
Patients and Methods: Eighty patients with confirmed *VHL* variants and RHs were retrospectively included and classified by variant type (missense vs. truncating) and variant location (α- vs. β-domain). Primary outcome was tumor regression; secondary outcomes were treatment-related complications and best-corrected visual acuity (BCVA) change. Structural models of VHL:ELOC and VHL:HIF1A complexes were generated using ColabFold, and predicted effects of missense variants on protein interactions were assessed with mCSM-PPI2.
Results: Tumor regression did not differ between truncating and missense variants (42.1% vs. 42.2%, p=0.995). α-Domain variants showed higher regression than β-domain variants (54.3% vs. 45.7%, p=0.017), although this was not independently predictive. Poorer baseline BCVA was associated with reduced regression (mean LogMAR: 0.55±0.98 vs. 0.16±0.41, p=0.025). Juxtapapillary tumors also showed lower regression rates (p=0.039). Cryotherapy (mean LogMAR +0.90±1.20, p=0.007) and anti-VEGF therapy (+1.01±1.24, p=0.032) were associated with visual decline. In silico analyses demonstrated that variants predicted to destabilize VHL–ELOC or VHL–HIF1A binding aligned with pathogenic classifications.
Conclusion: α-Domain *VHL* variants may be linked to better tumor regression, but baseline visual acuity and tumor location were stronger predictors. Combining molecular, structural, and clinical data supports a precision-medicine approach for managing ocular manifestations of VHL syndrome.

Genetic profiling from vitreous fluids as liquid biopsy for primary uveal melanoma: A proof-of-concept study

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Introduction: Uveal melanoma is an aggressive ocular malignancy. Molecular characterisation of primary tumours is crucial to identify those at risk of metastatic dissemination. Although tumour biopsies are being taken, liquid biopsies of ocular fluids may form a less invasive but relatively unexplored alternative. In this study, we aim to evaluate the DNA content of vitreous fluid from eyes with a uveal melanoma for molecular tumour information.
Materials and Methods: DNA was isolated from 65 vitreous fluid samples obtained from enucleated eyes with a uveal melanoma. Driver mutations and the copy numbers of chromosome 3p and 8q were studied using accustomed digital PCR assays. Our findings were compared to the molecular profile of matched primary tumours and to the clinicopathological tumour characteristics.
Results: Almost all vitreous fluids contained (traces of) DNA, but melanoma-cell derived DNA was detected in 45/65 samples (range 0.03%–94.4%) and was associated with a larger tumour prominence. Not all samples with melanoma-cell derived DNA harboured (analysable) secondary mutations or had sufficient statistical power to measure chromosomal copy numbers. Still, additional mutations were detected in 15/17 samples and copy numbers matched the primary tumour in 19/21 (chromosome 3p) and 18/20 samples (chromosome 8q). Collectively, a clinically-relevant molecular classification could be inferred in 29/65 patients.
Conclusion: This proof-of-concept study demonstrates the detectability of DNA in vitreous fluids from uveal melanoma patients, with melanoma-cell derived DNA in 69% of the samples. Prognostic genetic alterations could be identified in 45% of the patients. A follow-up study is needed to evaluate our approach in a prospective clinical context.

The role of chromosome 8q as an interdependent factor in prognosis of uveal melanoma patients

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Purpose: Previous studies have shown that gain of chromosome 8q in uveal melanoma (UM) is strongly correlated with shorter survival. In contrast, no correlation has been found between 8q gain and early-onset metastasis in *SF3B1*-mutated tumors. With a larger cohort, this study aims to establish the role of chromosome 8q gain as an interdependent prognostic factor.
Methods: This multicenter retrospective cohort study included mutation status of *GNAQ*, *GNA11*, *BAP1*, *SF3B1*, *EIF1AX*, *PLCB4* and *CYSLTR2*. Copy number was assessed using SNP-array, karyotyping or fluorescence in situ hybridization. Kaplan-Meier estimates were used for disease free survival analysis, and Cox proportional hazard regression was applied to identify the value of prognostic factors adjusted for age and sex.
Results: To date, 66 patients with SF3B1-mutated tumors have been included, of whom 17 (26%) developed metastatic disease. Gain of chromosome 8q was present in 56 tumors (84%). No association was found between 8q gain and disease-free survival after adjustment for age and sex ($p > 0.65$). Almost two-thirds of patients with a *BAP1* mutated tumor developed metastatic disease (143 of 234). Gain of chromosome 8q was present in 86% and was associated with decreased survival when four or more copies were present in the tumor (log-rank $p = 0.017$).
Conclusions: Gain of chromosome 8q is associated with decreased survival in patients with *BAP1*-mutated tumors, but it does not appear to influence survival outcomes in patients with *SF3B1*-mutated tumors.

Eye-preserving treatment and definitive eye retention in uveal melanoma

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Purpose: To evaluate eye-preserving treatment, secondary enucleation, and definitive eye retention in patients with uveal melanoma, and to assess the contribution of endoresection within the available treatment options.
Methods: All patients with primary uveal melanoma treated between 1995 and 2024 at Erasmus MC and The Rotterdam Eye Hospital were included. At Erasmus MC, fractionated stereotactic radiotherapy (fSRT) was introduced in 1999, endoresection in 2013, and proton therapy in 2020. For each five-year period, the proportion of patients treated with primary eye-preserving modalities (fSRT, proton therapy, and other eye-preserving treatments, and subsequent endoresection) was determined. Secondary enucleation was defined as enucleation following initial eye-preserving treatment and attributed to the period of the primary treatment. Definitive eye retention was defined as primary eye preservation minus secondary enucleation.
Results: The proportion of patients receiving primary eye-preserving treatment ranged between 48%–53% in earlier cohorts and increased to 83% in the most recent treatment period. The rate of secondary enucleation among initially eye-preserved eyes declined from 28% to 4%, resulting in an increase in definitive eye retention (as a proportion of all treated patients) from 38% (2000–2004 cohort) to 80% (2020–2024 cohort). Following its introduction, endoresection contributed an increasing share of definitive eye retention across successive five-year cohorts, resulting in 3, 14, and 19 additional eyes preserved in the 2010–2014, 2015–2019, and 2020–2024 cohorts, respectively. Overall, 49 endoresections were performed; 10 eyes required enucleation, resulting in 39 preserved eyes, predominantly in patients with medium-sized and selected large tumors.
Conclusions: Eye-preserving treatment strategies are associated with substantial definitive eye retention in uveal melanoma. Endoresection provides a clear additional contribution to eye retention alongside radiotherapeutic treatment option.

Radiological surveillance in BAP1-tumor predisposition syndrome patients using MRI-chest-abdomen

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Introduction: Patients with BAP1-tumor predisposition syndrome (BAP1-TPDS) have a life-time risk of 85% to develop ≥1 malignancy. Thus, the use of MRI-chest-abdomen (MRI-CA) imaging is recommended for regular cancer screening. However, evidence evaluating the yield is lacking. We aim to assess the clinical utility of MRI-CA in BAP1-TPDS patients at baseline and follow-up.
Patients and methods: We followed 66 patients with BAP1-TPDS seen at the Leiden University Medical Center from July 2017 till September 2025 prospectively.
Results: Out of the 66 patients with BAP1-TPDS, 42 were female (63.6%). In total 181 MRI screenings were performed, with a mean of 2.7 ± 1.2 screenings per patient. The median age at baseline was 49 years (range 25–74 years) and 52 years (range 30–74 years) for subsequent screenings. Initial screening led to an imaging or invasive intervention in 15.2% of scans, compared to

5.2% of subsequent screening. Three (4.5%) cancers were detected at baseline MRI- CA, and one (0.9%) at subsequent screening. At baseline the sensitivity and specificity MRI-CA were 60% and 88.5%, respectively, with a positive predictive value (PPV) of 30%, negative predictive value (NPV) of 96.4%, false-positive rate (FPR) of 11.5% and false-negative rate (FNR) of 40%. The subsequent screenings showed a sensitivity of 50%, specificity of 95.8%, PPV of 16.7%, NPV of 99.1%, FPR of 4.4% and a FNR of 50%.

Conclusions: These findings provide important data for counseling BAP1-TPDS patients and show that MRI-CA is an effective screening tool, comparable to MRI screening in other tumor predisposition syndromes.

Multifocal intraocular lens placement in the sulcus during refractive lens exchange: A case series

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Purpose: Posterior capsular rupture presents a challenge in refractive lens procedures, as most multifocal IOLs are not designed for sulcus placement. This study evaluates refractive outcomes of a multifocal IOL that was designed to be placed in the bag but can be placed in the sulcus due to haptic angulation. Here we report the results of 5 eyes in which this lens was placed in the sulcus.

Setting: Private refractive surgery clinic, the Netherlands.

Methods: This diffractive, one-piece hydrophilic IOL with 5-degree posterior angulation serves as both a primary and backup option. Between 2008 and 2023, five eyes from four male patients (aged 36–59) required sulcus placement following RLE. Two eyes received the Intensity SL MF IOL, and three received the SeeLens MF IOL. Indications included posterior capsular rupture (three eyes) and capsular contraction syndrome in nanophthalmos (two eyes). Sulcus placement was performed during primary surgery in two eyes, as a secondary procedure in one, and to correct hypermetropization in two eyes with capsular contraction.

Results: Postoperative uncorrected and corrected distance visual acuities averaged 0.7 and 1.1, respectively. Mean postoperative refraction was S -0.25 C −0.45. One patient developed retinal detachment at eight months, which was successfully treated. All patients were satisfied post-repositioning.

Conclusions: Sulcus placement of the SeeLens MF or Intensity SL IOL is a viable option when necessary.

Development and Validation of the Global Spectacle Independence Questionnaire: A brief patient-reported outcome measure with U.S. population norms

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Purpose: To develop and validate the Global Spectacle Independence Questionnaire (GSIQ), a patient-reported outcome measure of spectacle independence (SI), and to establish U.S. national normative values for SI.

Methods: Instrument development followed FDA and ISPOR guidance. Concept elicitation and cognitive debriefing interviews were conducted among U.S. adults (N=36). Psychometric evaluation data were collected via an online survey representative of the U.S. adult population (N=960). Analyses included internal consistency, test-retest reliability, exploratory factor analysis, convergent validity relative to NEI-RQL-42, known-groups validity, and logistic regression. The primary endpoint, GSIQ-SI, was defined as responding “None of the time” to all three core items – need for vision correction, wearing vision correction, and straining/squinting without correction – during the past 7 days.

Results: Qualitative interviews confirmed the three core concepts and demonstrated concept saturation. The final GSIQ contains three core items and one optional digital-device item. Factor analysis supported unidimensionality (ICC=0.87). Convergent validity with NEI-RQL subscales was moderate to strong, and known-groups analyses and logistic regression supported construct validity. In the weighted U.S. normative sample, 18.6% of adults met the criterion for complete SI.

Conclusions: The GSIQ is a concise and reliable instrument with strong evidence of validity. Its dichotomous primary endpoint provides a clear and practical measure of complete spectacle independence suitable for clinical trials, regulatory labeling claims, and real-world evidence studies.

Differences in peripheral contrast sensitivity and peripheral optical errors between phakic and pseudophakic eyes

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Purpose: Peripheral contrast sensitivity (CS) is crucial for functional vision such as navigation and object detection. Although cataract surgery aims at restoring central vision, peripheral optical quality is altered in pseudophakic eyes. This study compared peripheral CS between phakic and pseudophakic participants

and examined how peripheral refractive errors (PREs) contribute to these differences.

Methods: Fourty-eight participants (24 pseudophakic) with monofocal IOLs completed two monocular runs of a CS test at 20° nasal visual field eccentricity. Participants reported the presence of a Gabor patch (1.6°, 3 cpd;500 ms) in a detection task, with contrast controlled by a Bayesian adaptive staircase. PREs were measured with the Grand Seiko WAM-5500 and used to derive blur and astigmatism at seven locations spanning 30° nasal to 30° temporal.

Results: Higher blur ($r=-0.53$, $p<0.001$) and astigmatism ($r=-0.45$, $p<0.001$) were associated with lower CS. Phakic participants showed higher peripheral CS (1.53 ± 0.176) than pseudophakic participants (1.30 ± 0.219, $F=15.33$, $p<0.001$). CS performance across runs was highly consistent ($r=0.92$, $p<0.001$). Pseudophakic eyes exhibited greater blur at 20° nasal (3.818 ± 1.87D) than phakic eyes (1.437 ± 0.955D, $p<0.001$). Astigmatism was higher in pseudophakic eyes (3.30 ± 1.70D) than phakic eyes (1.71 ± 0.86D, $p=0.001$).

Conclusions: PREs have a substantial impact on peripheral CS. Differences in peripheral optics between phakic and pseudophakic eyes lead to measurable reductions in peripheral visual function in individuals with pseudophakia following cataract surgery.

Scleral lens power surprise after cataract surgery in patients with irregular corneas

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Purpose: Intraocular lens (IOL) calculation for cataract surgery is complicated in irregular corneas due to various reasons. In scleral lens wear, the (negative) tear lens power must also be accounted for to achieve a satisfied outcome. This study describes significant differences in scleral lens power before and after cataract surgery in patients with irregular corneas.

Methods: This retrospective study analyzed data from scleral lens patients with irregular corneas who underwent cataract surgery and subsequently required a marked change in scleral lens power. Demographics, scleral lens fitting and power (spherical equivalent [SE], diopters [D]) before and after surgery, and the indication for scleral lens wear were evaluated.

Results: Of 25 patients (48% female), scleral lenses of 33 eyes were analysed. Median age at the time of the surgery was 68 years (range 39 to 76) and 51.5% had surgery in OD. The scleral lens indication was keratoconus in 79%, keratoplasty in 15% and irregular astigmatism in 6%. Mean scleral lens SE changed from +0.53D (SD 2.30, range −3 to +7) pre-op to +9.8D (SD 3.9, range +2.5 to +21.5) post-op, $p<0.001$.

Conclusions: Despite advances in IOL power calculations for irregular corneas, scleral lenses showed a remarkable pre- to post hyperopic shift after cataract surgery, which may lead to anisometropia, aniseikonia and hypoxia (due to increased lens thickness). Therefore, when calculating IOL power in patients with irregular corneas who are expected to remain dependent on their lenses after surgery, consideration of the scleral lens (power) would be valuable.

Linking patient-reported and clinical outcomes in cataract surgery: Insights from a three-year cohort with integrated predictive modelling

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Introduction: The interactions between clinical outcome measures (COMs), patient-reported experience measure (PREM) and patient-reported outcome measures (PROMs) is of vital importance to understand the reported outcomes of cataract surgery. However, these interactions are complex and not fully elucidated. Our study investigates the association between postoperative best corrected visual acuity (BCVA), PREM and Catquest 9-SF in cataract surgery.

Patients and methods: All patients who received cataract surgery were included at Xpert Clinics Oogzorg Zeist between 2021 and 2024. Patient demographics and BCVA were collected during the initial outpatient visit and a postoperative follow-up of 4–6 weeks. The pre-operative PROM was sent by email after their initial outpatient visit and the post-operative PROM and PREM were sent 3 months after cataract surgery. The distribution between BCVA gain, PROMs and PREM results was visualized in violin plots. The influence of patient characteristics, baseline factors (PROMs and PREM) and process factors on the postoperative vision was measured via hierarchical regression.

Results: Of 4154 included patients, 3092 patients filled in the Catquest 9-SF and 2113 patients filled in the PREM, which showed improved visual outcomes and high satisfaction after surgery. The full predictive model explained 33% of the variance in postoperative visual acuity ($R^2=0.33$).

Conclusion: PROMs correlate with, but are not identical to clinical outcomes. Although the Catquest 9-SF and PREM do not fully capture a patient's health status, they do provide complementary insights beyond clinical outcomes. A more elaborate PROMs measurement track should be integrated in value-based cataract care.

Correcting astigmatism using toric intraocular lenses during cataract surgery

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Purpose: Astigmatism is one of the most common refractive conditions, with a large proportion of patients undergoing cataract surgery presenting with 1.0 D or more. Toric IOLs are the standard of care for astigmatism correction during cataract surgery. This overview of reviews synthesizes current evidence on the use of toric IOLs, focusing on clinical indications, surgical outcomes, and emerging technologies.

Methods: The following databases were searched from inception to 2 June 2025: MEDLINE, Embase, the Cochrane Library (including CENTRAL and the Cochrane Database of Systematic Reviews), KSR Evidence, and the Trip Database. Keywords included astigmatism, toric intraocular lenses, cataract surgery, intraocular lens implantatio, and postoperative outcomes. Included studies were systematic and narrative review, meta-analyses, and clinical guidelines related to the use of toric IOLs in cataract surgery.

Results: Eighty-five reviews were included. This overview summarizes key topics including the prevalence and impact of astigmatism, toric IOL technology, preoperative assessment, power calculation methods, surgical considerations, and postoperative outcomes such as rotational stability and surgically induced astigmatism. Reporting standards, patient-reported outcome measures, adverse events, and considerations in special populations are also discussed. Knowledge gaps and debated areas are highlighted.

Conclusions: Toric IOLs improve uncorrected visual acuity and refractive outcomes in cataract patients with significant astigmatism. Advances in diagnostics, lens design, calculation formulas, and intraoperative tools continue to enhance accuracy and patient satisfaction. Further research is needed to address unresolved questions and support best practices in diverse clinical settings.

Cross facial corneal neurotization, 3 cases

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Introduction: Corneal neurotization is a technique to reinnervate an anesthetic cornea by connecting it directly or via an autologous nerve transplant to a healthy sensory nerve in order to treat neurotropic keratopathy.

Methods: Between 2019 and 2024 we performed a neurotization procedure in 3 patients, connecting the left cornea with the right supra-orbital- and or supra trochlear nerve via an autologous sural nerve transplantation.

Results: The first patient was a 54-year-old male with a neurotrophic corneal ulcer OS after surgery for an acoustic neuroma complicated by a failure of the left nV, nVII and nIX. After neurotization the corneal ulcer healed and there was a minimal subjective increase in corneal sensation. The second patient was an 8-year-old boy with a neurotrophic corneal ulcer OS of unknown origin. After neurotization the ulcer healed but the Cochet Bonnet measured corneal sensibility did not increase. The third patient was a 7-year-old girl with a variation in the ZBTB34 gen with psychomotor retardation, epilepsy and a neurotrophic corneal ulcer OS. After neurotization the corneal ulcer healed and the sensibility increase from 0 to 50 mm.

Conclusion: The corneal neurotization procedure in these three patients showed a satisfactory healing response in regard to the neurotrophic ulcers but only one patient showed a measurable sensory recovery.

CorNeat keratoprosthesis with an electrospun Flange

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Purpose: To report the outcomes of implantation of a novel keratoprosthesis comprised of a PMMA dome, with an electrospun flange (CorNeat, USA).

Methods: Retrospective case-series. In 2 eyes of 2 patients a CorNeat Keratoprosthesis was implanted. The indication for implantation was corneal blindness with recurrent corneal graft failure, and complex ocular comorbidity. These patients are included in a multicenter international phase 1 study. Treatment is according to study protocol. Outcome measures were visual acuity, visual analog pain score (VAS), and complications.

Results: In 2 uneventful implantation was done. In case 1 visual acuity improved from logMAR 2.0 to 0.2, and case 2 from logMAR 2 to 0.6. VAS scores improved in both. Complications: In patient 1 conjunctival retraction was seen, both patients had episodes of CME.

Conclusion: Keratoprosthesis is a notoriously difficult treatment modality reserved as a last resort. While having the potential of restoring useful vision, it is also prone to serious complications. With more traditional Kpro keratoprostheses integration of the implant into the donor site was one of the frequent causes of failure. The development of an electrospun flange that can integrate with the host tissue is thought to make the difference in this implant. Short and limited experience with this new implant have shown promising results, but still, we need to learn more about the implant behavior and the long-term management.In corneal blindness with poor prognosis for re-keratoplasty keratoprosthesis is a complex treatment option. 2 cases of implantation of a CorNeat keratoprosthesis with electrospun flanges are presented. The CorNeat's novelty is the integration of a central PMMA dome attached to an electrospun flange, that is

porous, and designed to integrate with the patients tissue and prevent the point of failure of earlier keratoprostheses designs. Patients are participating in a multicenter internation Phase 1 trial. In both patients' improvement in visual acuity (from LogMAR 2.0 to LogMar 0.2 and 0.6 respectively) and pain scores were noted. Complications encountered were conjunctival retraction, and CME. Duration of follow up now is between 3-10 months. Short term results with this novel keratoprosthesis look promising, however more and longer follow up is necessary. Treatment with a keratoprosthesis potentially restores vision, but is an intensive treatment option.

Improving visual performance with soft keratoconus contact lenses

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Introduction: Scleral lenses are often considered the best optical aid for restoring visual performance in patients with keratoconus. A study of 100 keratoconus patients, corrected with soft keratoconus lenses, offers new perspectives.
Methods: The first 100 keratoconus patients who received a soft keratoconus lens fitting or were scheduled for follow-up in 2025 were selected. Corneal parameters and higher-order aberrations were assessed with and without correction (s.c.) using Placido ring topography and ray tracing aberrometry (I-Tracey). Achieved visual acuity with soft keratoconus lenses were related to K-steep and the reduction of higher-order aberrations. We also included patient experiences with other worn types of contact lenses.
Results: No experience yet with contact lenses ($n=22$), scleral lenses ($n=40$), corneal lenses ($n=29$), soft contact lenses ($n=17$). The mean visual acuity s.c. was found to be V0.38 ($n=107$), with spectacle correction V0.75 ($n=104$), with corneal or scleral lenses V0.92 ($n=40$), with soft keratoconus lenses V1.08 ($n=192$). The relationship between achieved visual acuity and mean K-steep was found to be: V1.5 (44.62D), V1.2 (46.08D), V1.0 (48.91D), V0.8 (47.91). Higher order aberrations were reduced by an average of 55% compared to s.c.
Discussion and conclusions: Most patients had previous contact lens experience. The highest average visual acuity, per lens type, was achieved by soft keratoconus contact lenses. K-steep has a significant correlation with the visual acuity by a soft keratoconus lens. Higher-order aberrations significantly reduced by the correction of soft keratoconus lenses.

Long-term results of eyes with a corneal transplantation and a glaucoma drainage device: A Dutch retrospective multicenter study

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Purpose: To evaluate clinical outcomes in eyes with keratoplasty and glaucoma drainage device (GDD) implantation.
Methods: This retrospective case series included 137 eyes from two tertiary centers in the Netherlands, treated between October 2002 and March 2021. Patients were divided into two groups: Prior GDD (GDD before keratoplasty) and Subsequent GDD (GDD after keratoplasty). The primary outcome measure was graft survival. Secondary outcomes included intraocular pressure (IOP), IOP-lowering medication use, best-corrected visual acuity (BCVA), and postoperative complications. Statistical analyses included Kaplan-Meier, Cox regression, and linear mixed models.
Results: Graft survival was significantly lower in the Prior GDD group compared to the Subsequent GDD group at 2 years (72% vs. 95%) and 5 years (47% vs. 74%) ($p<0.001$). IOP success at 2 years was 94% in the Prior GDD and 97% in the Subsequent GDD group. Trends in IOP, IOP-lowering medication, and BCVA differed significantly between groups (all $p<0.001$). In the Prior GDD group, IOP and medication use remained stable, while BCVA improved significantly from 1.76 to 1.01 logMAR at 3 months ($p<0.001$). In the Subsequent GDD group, IOP decreased from 28.3 to 13.3 mmHg at 3 months ($p<0.001$), and medication use dropped from 4 to 2 agents ($p<0.001$), while BCVA remained stable. Postoperative complications occurred in 25% of eyes, with no significant difference between groups.
Conclusions: Graft survival was lower in eyes with Prior GDD, but both groups achieved favorable IOP control and visual outcomes with low complication rates. These findings underscore the importance of surgical sequencing to optimize outcomes.

Outcomes of visual rehabilitation with contact lenses after corneal ulcer healing

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Introduction: Visual rehabilitation after a corneal ulcer can be challenging. Contact lenses (CL) may offer improvement, yet evidence on visual outcomes and patient tolerance in this population is limited. This study aims to assess the visual outcomes and patient-reported tolerance of CL wear in individuals with complete healing following a corneal ulcer.
Patients and Methods: This retrospective descriptive study included patients between January 2006 and January 2025 with a diagnosed corneal ulcer who, after complete healing, were referred for CL fitting to improve visual acuity (VA). Data on CL type, VA prior to fitting, VA with the prescribed CL, ulcer location and patient-reported complaints were obtained.
Results: 19 patients were included: 15 were fitted with a corneal rigid gas permeable (RGP) lens and 4 with a scleral lens. Median VA improved from 0.40 [0.78] to 0.10 [0.12] logMAR with corneal RGP lenses and from 0.50 [1.00] to 0.31 [0.59] logMAR with scleral lenses. VA improved after CL fitting regardless of whether the ulcer was centrally or paracentrally located. 7 patients reported complaints during CL wear, and 3 eventually discontinued use.
Conclusions: Both corneal RGP and scleral lenses provide clinically relevant improvement in VA after complete healing of a corneal ulcer. Only a minority of patients discontinued CL wear, indicating good tolerance in this population. Referral to an experienced practitioner, individualized CL selection, and realistic patient counseling appear essential for optimizing outcomes. These findings support the use of CL as an effective option for visual rehabilitation following corneal ulceration.

A translational pipeline for autologous conjunctival organoid grafts toward first-in-human application

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Introduction: Damage or loss of conjunctival tissue can lead to chronic irritation, scarring, tear film instability and ocular surface failure. Current surgical approaches often lack sufficient donor tissue or fail to restore native conjunctival architecture. We developed an organoid-based, autologous strategy to generate transplant-ready conjunctival grafts and advanced this platform toward GMP-compliant clinical translation.
Materials and Methods: Human conjunctival biopsies were expanded as epithelial organoids in a dome-based 3D culture system. Organoids were dissociated into single cells and seeded onto fibrin–thrombin gels, forming confluent epithelial sheets within four days. For translation, the research process was converted to a GMP-compatible flow with IMPD preparation, validated QC assays, and defined release criteria; a specialized GMP facility has been engaged for clinical-grade manufacturing.
Results: The organoids provided a renewable source of patient-specific conjunctival epithelium. Both isolated cells and mature organoids can be cryopreserved and later re-expanded, enabling long-term patient-specific tissue storage and avoiding additional biopsies in cases of recurrent disease. The grafts were suitable for surgical manipulation and demonstrated robust regenerative performance in a pre-clinical in vivo model. These findings confirmed the biological efficacy of the grafts and supported their progression toward clinical translation. In parallel, the complete R&D-to-GMP conversion was successfully finalized, including validated quality-control assays, established release criteria, and selection of a GMP facility for clinical-grade manufacturing.
Conclusion: This work establishes a GMP-ready pipeline for autologous conjunctival organoid grafts, including cryopreservation for future use, supporting initiation of a first-in-human study for recurrent pterygium in early 2027.

Mapping keratoconus genetics using classic and endophenotype-based GWAS approaches reveals novel loci

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Introduction: Keratoconus is a progressive corneal disorder that typically emerges in early adulthood, causing structural deformation and visual impairment. Despite a relatively strong genetic component, the genetic basis of keratoconus remains largely unresolved. This study aimed to advance our understanding of keratoconus genetics using a multi-stage genome-wide association study (GWAS) framework.

Methods: We employed a three-stage design which integrated quantitative corneal endophenotypes with case-control analyses. In stage-I, 33 Pentacam-derived corneal parameters were assessed in 16,464 individuals of European ancestry across five cohorts to prioritize the most heritable and genetically relevant keratoconus endophenotypes. Stage-II employed a multi-trait GWAS of prioritized traits, followed by evaluation in case-control cohorts. Stage-III comprised a comprehensive case-control meta-analysis (6372 cases/593,541 controls). Linear and logistic models were used for association testing, with summary statistics combined using fixed-effect inverse variance and multi-trait meta-analysis. Gene-set enrichment and tissue-expression analyses explored biological pathways. Significance thresholds were set at 5.0×10^{-8} or a Bonferroni-adjusted p-value, as appropriate.

Results: Four endophenotypes met prioritization criteria: posterior corneal asphericity, posterior best-fit sphere radius, central corneal volume (5 mm), and maximum anterior keratometry. Multi-trait analysis identified 80 independent endophenotype-associated loci. Of those, 25 loci (five novel) were replicated in keratoconus cohorts. The final case-control meta-analysis yielded 51 loci significantly associated with keratoconus (eight novel). Gene-set enrichment implicated extracellular matrix organization, glucosylceramidase regulation, and cell differentiation.

Conclusion: This work defines 11 novel loci and key pathways in keratoconus, demonstrating the power of phenotype-integrated GWAS for mapping disease biology, and laying the ground for improved genetic risk profiling.

Randomised trial comparing visual outcomes and quality of life after UT-DSAEK and DMEK in patients with Fuchs endothelial dystrophy

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Purpose: Posterior lamellar keratoplasty techniques are currently the preferred option to restore vision in advanced cases of Fuchs' endothelial dystrophy (FED). Former studies detected small differences in visual acuity between Ultrathin Descemet Stripping Automated Endothelial Keratoplasty (UT-DSAEK) and Descemet Membrane Endothelial Keratoplasty (DMEK), while patient reported outcomes failed to detect a difference between the two techniques. This study aimed to compare visual outcomes and patient-reported outcomes after UT-DSAEK and DMEK.

Methods: In this randomized controlled trial, pseudophakic eyes of patients with FED without relevant ocular pathology were randomized for UT-DSAEK or DMEK surgery. The primary endpoint parameter

was best-corrected visual acuity (BCVA) at 12 months. Subjective improvement was measured by means of VFQ-25 and EQ-5D questionnaires. Complication parameters including post-operative intraocular pressure, rebubbling rate, and cystoid macula edema were assessed. Statistical analysis included linear regression models.
Results: A total of 81 eyes of 81 patients were eligible for the 12-month follow-up. Baseline characteristics were comparable between the two groups, including mean age (70.8 ± 6.6 years in UT-DSAEK; 73.5 ± 4.6 in DMEK) and baseline LogMAR BCVA (0.36 ± 0.26 vs. 0.38 ± 0.32). After adjustment for baseline BCVA, Regression analysis demonstrated that DMEK resulted in significantly better 12-month BCVA (0.082 LogMAR) compared with UT-DSAEK (0.185 LogMAR; $\beta = -0.1038$, 95% CI [−0.167, −0.041], $p = 0.002$).
Conclusions: In this randomised trial performed in patients with FED without relevant ocular pathology, visual acuity was significantly better in the DMEK group as compared to eyes undergoing UT-DSAEK. Complication rates were comparable in both groups.

Feasibility assessment of automated corneal endothelium quantification using CELDA in donor corneas

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Purpose: Endothelial cell density (ECD) is essential for determining the quality of donor corneal tissue. However, current assessment methods rely on manual counting and are subject to inter-operator variability. CELDA is a GUI (Graphical User Interface) that deploys a deep-learning model trained on specular microscopy images from patients' corneal endothelium. This feasibility study evaluates CELDA's performance for automated ECD determination in *ex vivo* donor corneas at ETB-BISLIFE Eye Bank in Haarlem, The Netherlands.
Materials and Methods: CELDA was tested on 34 heterogeneously focused images, acquired from six donor corneas, to determine which image characteristics it handles best. Scale and blur preprocessing approaches were tested to enhance robustness. Twelve corneas were prospectively imaged at one central and four peri-central locations targeting the identified image characteristics. CELDA's outputs were compared with manual ECD counts using means, Lin's concordance correlation and Bland-Altman analyses.
Results: CELDA performed well on images with low-intensity backgrounds, high-intensity cell bodies, and reduced detail. Preprocessing with a 0.3 scaling factor improved robustness, with cells recognized in 93% of prospectively acquired images; cells were not recognized in four images from a single cornea due to insufficient quality. CELDA based its ECD calculation on more cells per cornea than manual counting (2475 [744–4390] vs 132 [118–140]) with an increasing difference between the methods at higher ECDs (mean difference 182 cells/mm^2; 95% CI −167.3 to 533.1), with decreased agreement toward the periphery ($r_c = 0.77$ for central vs 0.64 for full cornea). CELDA showed promising intra-corneal agreement ($r_c = 0.64$; 0.94 after outlier removal).
Conclusions: CELDA shows great potential for automated ECD quantification in donor corneas. However, standardisation of image acquisition, model retraining, and extensive validation are required before implementation of this fully automatic approach at Eye Banks using light microscopy.

Lumican distribution in endothelium and stroma of donor corneas with and without Guttae

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Purpose: Guttae are extracellular matrix (ECM) protrusions starting in early stages of Fuchs endothelial corneal dystrophy (FECD). Lumican is a crucial ECM proteoglycan in the corneal stroma, vital for maintaining its transparency by regulating collagen fibril assembly and is involved in immune responses during inflammation. We investigated the spatial organization of Lumican in endothelium and stroma of guttae vs healthy corneas of human donor eyes to analyse the potential difference and anatomical-functional correlation.
Methods: Immunohistochemistry and confocal imaging were performed on 4% paraformaldehyde fixed flat-mounted central bottoms from human donor corneas (without or with guttae) for Lumican and Neural Cell Adhesion Molecule (NCAM) as a comparable ECM fluorescent signal.
Results: While healthy corneas demonstrated distinct spatial patterns for Lumican expressed mainly in intertwined web-like distribution in stroma in relation to keratocytes and NCAM, guttae corneas showed altered Lumican morphology appearing like sporadic aggregations pushing cells in endothelium. Interestingly, in 3D volume viewer of approximately 50–100 μm z-stacks, in healthy corneas, Lumican appeared as repeated parallel posterior-to-anterior cords compared to altered Lumican signal in guttae endothelium and stroma.
Conclusions: This study examined ECM Lumican glycoprotein distribution, revealing distinct spatial and vertical appearance across healthy corneas which is altered in cornea guttata. These findings suggest that guttae could

be related to altered Lumican distribution in both endothelium and stroma.

Densitometry prediction of secondary corneal decompensation (CD) after phacoemulsification only (PO) in Fuchs endothelial dystrophy (FED)

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Purpose: To evaluate Pentacam central Densitometry performance in predicting corneal decompensation after PO, relative to other methods for assessment of (sub) clinical edema.
Methods: Between 2021 and 2023, 75 consecutive patients with FED and cataract in a single centre prospective cohort were evalutated with a cut-off of >25.9 Pentacam central (0–2 mm) total Densitometry aiding clinical evaluation, for predicting CD after PO. Postoperatively, secondary EKs (SEK) within 1 year, as well as the predictive performances of preoperative Patel Pentacam criteria (PPC) and central corneal pachymetry used for assessing subclinical corneal edema, were evaluated.
Results: Regarding eyes with evident clinical corneal edema and/or high Densitometry, 1 patient refused (T)EK and was removed from this analyis, and 27 patients underwent triple endothelial keratoplasty (TEK). Pachymetry and PPC classified all eyes correctly, whereas Densitometry underclassified 3/27, with values <25.9. The remaining 47 eyes underwent primary PO; from which 2 eyes needed SEK within one year. Preoperative Densitometry >25.9 predicted and correctly classified both SEKs; and overclassified 2/47 other eyes. In contrast, PPC underclassified - i.e. failed to predict - both SEKs, and overclassified 7/47 eyes; whereas pachymetry underclassified 1/47 and overclassified 6/47 eyes.
Conclusions: In our cohort, for the advice PO in patients with FED, Densitometry >25.9 predicted SEK better than pachymetry, which performed better again than PPC.

Functional validation of iPSC-derived corneal endothelial cell products in preclinical rabbit transplantation

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The differentiation of iPSCs into induced corneal endothelial cells (iCEnCs) offers a promising strategy to address the global shortage of donor corneas for transplantation. Our optimized differentiation protocol consistently generates CEnC-like cells with high efficiency. Product characterization by immunofluorescence and flow cytometry shows expression of iCEnC markers, including ZO-1, ATP1A1, and CD166, while confirming minimal expression of off-target markers, such as SOX2 and CD44. Differentiation efficiency yields high CEnC-like cells with cell-cell contact and expression of CD166 and N-cadherin. To evaluate the therapeutic potential of these differentiated CEnCs, we are establishing a rabbit model for preclinical assessment of corneal endothelial function and graft integration. In this model, we remove the native corneal endothelium from rabbit eyes and inject iCEnCs into the anterior chamber to replace the damaged endothelium. We measure corneal thickness over time as a functional readout; successful cell engraftment is expected to result in progressive corneal thickness reduction as the iCEnCs restore barrier integrity and fluid homeostasis. Following the in vivo observation period, we harvest and analyze corneal tissues to assess cell maturation and phenotypic changes in response to the in vivo environment. Given the fetal-like characteristics of our differentiated iCEnCs, this tissue analysis provides critical insights into how the cells respond to physiological cues and whether they acquire more mature, adult-like endothelial properties in vivo. Our iPSC-derived CEnC product generation and preclinical rabbit model validation provide critical evidence for the therapeutic potential of cell-based regenerative therapies for corneal endothelial dysfunction, supporting the pathway toward clinical application.

Dupilumab-associated ocular surface disease in atopic dermatitis: Results from the BioDay registry

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Introduction: Dupilumab-associated ocular surface disease (DAOSD) is a frequently reported side effect in atopic dermatitis (AD) patients treated with dupilumab. This study aimed to investigate the frequency and severity of DAOSD and the effect of dupilumab on conjunctival goblet cells (GCs) in a large prospective real-world cohort.
Patients and Methods: This prospective study included moderate-to-severe AD patients from the BioDay registry

treated with dupilumab between February 2020 and January 2025 at the UMC Utrecht. Ophthalmological and dermatological examinations were performed at baseline (start of dupilumab), week 4, and week 28. Ocular surface disease (OSD) severity was assessed using the Utrecht Ophthalmic Inflammatory and Allergic disease (UTOPIA) score. DAOSD was defined as a ≥3-point increase from baseline. Conjunctival impression cytology was performed to study the quantity and function of conjunctival GCs.

Results: OSD was present in 94.0% (n=141/150) of patients at baseline, while only 60.0% (n=90/150) of patients reported ocular symptoms. During 28 weeks of dupilumab treatment, 30.7% (n=46/150) of patients developed DAOSD. At week 4 and week 28, 56.7% (n=85/150) and 64.7% (n=97/150) of patients regularly used ophthalmic medication, respectively. GC numbers remained stable between baseline and week 28, while Mucin 5AC (MUC5AC) production in Cytokeratin 19-CD45-MUC5AC+ cells significantly decreased.

Conclusion: This study highlights the high prevalence of OSD in moderate-to-severe AD patients before dupilumab treatment. DAOSD was observed in 30.7% of patients, despite the potential protective effect of ophthalmic treatment. While conjunctival GC numbers remain stable but low, dupilumab seems to impair GC function.

Pre- and postoperative reflection of fear and postoperative satisfaction in DMEK under general anesthesia versus sub-tenon anesthesia

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Purpose: This study explored differences in pre- and postoperative surgical fear and postoperative satisfaction in patients undergoing Descemet membrane endothelial keratoplasty (DMEK) under general anesthesia (GA) compared to sub-tenon anesthesia (STA).

Methods: In this prospective observational cohort study, 50 patients undergoing DMEK were included (21 GA group, 29 STA group). Preoperative fear was assessed using the Surgical Fear Questionnaire (VVO) two weeks prior to surgery (t=–1). Postoperative satisfaction, perceived recovery and postoperative reflection of fear were measured at four timepoints: directly after surgery (t=0), one week (t=1), one month (t=2) and three months (t=3) using the Global Perceived Effect (GPE/GEE) and postoperative VVO. Group differences were analyzed using independent t-tests and Fisher-Freeman-Halton exact tests.

Results: Total VVO scores showed no significant differences between the two anesthesia groups before nor at any time point after DMEK (all $p>0.05$). Postoperative GPE/GEE outcomes—perceived recovery, treatment satisfaction, and symptom change—were comparable between the groups without statistically significant differences. Both groups demonstrated marked improvement (100% vs 90%) and high satisfaction (100% vs 90%) after three months. Patients in the GA group more frequently indicated they would choose the same anesthesia again (100% vs 76.5% at $t=2$ $p<0.05$).

Conclusions: Pre- and postoperative fear, perceived recovery and overall satisfaction after DMEK did not differ between the GA and STA group. While patients receiving GA would more often prefer the same anesthesia method again, STA appears to be a well-tolerated, patient-acceptable alternative to GA for DMEK.

DMEK endothelial cell density underestimated by specular microscopy

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Purpose: DMEK endothelial cell densities (ECDs) using inverted light microscopy in the eye bank preoperatively and post mortem, were compared to ECDs obtained by clinical specular microscopy in the same DMEK corneas, to determine discrepancies in laboratory-based and in vivo imaging.

Methods: Eighteen DMEK corneas of 14 patients were donated post-mortem. All corneas had clinical specular microscopy at regular follow-up intervals from 6 months up to 6 years postoperatively. ECD counts in the eye bank were compared to the specular microscopy ECDs as well as to post-mortem ECDs, using the same inverted light microscope, camera and software.

Results: Compared to the preoperative ECDs in the eye bank, apparent ECD loss by specular microscopy at 6 months averaged 38 ± 18%, and steadily decreased over time. However, post-mortem ECDs only showed a 25 ± 22% decrease compared to preoperative ECDs.

Conclusions: An apparent 38% ECD loss in DMEK eyes in the first months after surgery may in part be explained by ECD underestimation with clinical specular microscopy, since preoperative to post-mortem ECD counts in the eye bank using the same light microscopy set-up only showed a 25% drop in ECD.

Pathological mechanisms in Fuchs endothelial corneal dystrophy and myotonic dystrophy type 1: More than meets the eye

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Introduction: Fuchs endothelial corneal dystrophy (FECD) and myotonic dystrophy type 1 (DM1) are trinucleotide repeat (TNR) disorders caused by non-coding CTG expansions in *TCF4* and *DMPK*, respectively. Despite this shared genetic feature, the two diseases differ markedly in clinical phenotype, tissue involvement, and disease course.
Methods: Experimental and clinical studies addressing molecular mechanisms in FECD and DM1 were evaluated in conjunction, focusing on repeat instability, RNA-mediated toxicity, dysregulation of RNA-binding proteins, alternative splicing, repeat-associated non-AUG (RAN) translation, and potential cis- and trans-modifying factors.
Results: Both disorders show somatic instability of CTG repeats and RNA-mediated toxicity characterized by nuclear RNA foci formation and sequestration of RNA-binding proteins, including MBNL proteins. In DM1, these mechanisms result in a multisystem disorder with widespread splicing abnormalities. In contrast, FECD pathology is largely restricted to the corneal endothelium. Evidence in FECD further suggests cornea-specific repeat instability, potential RAN translation products, and altered expression of *TCF4* isoforms. The contribution of DNA repair pathways as trans-modifiers appears to be locus- and cell type-dependent.
Conclusion: Although FECD and DM1 share key molecular features as TNR disorders, distinct locus- and tissue-specific mechanisms underlie their divergent clinical manifestations. These insights contribute to a better understanding of FECD pathogenesis and highlight directions for future mechanistic and translational research.

DMEK in vitrectomized eyes: Modified surgical techniques and clinical outcomes

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Introduction: Descemet Membrane Endothelial Keratoplasty (DMEK) in vitrectomized eyes is surgically demanding due to the lack of vitreous support, which leads to a deep anterior chamber and limited graft control. The presence of silicone oil tamponade may further complicate graft manipulation and requires adapted surgical techniques.
Materials and Methods: This presentation reviews surgical strategies for performing DMEK in vitrectomized eyes. Two techniques are illustrated using intraoperative surgical videos. The application of an anterior chamber maintainer is demonstrated in a case involving silicone oil tamponade and a prepupillary iris-claw IOL. A retrospective evaluation of five cases was conducted, assessing endothelial cell loss, rebubbling rate, and change in visual acuity at six months.
Results: Complete graft attachment was achieved in all cases, and no rebubbling was required. Mean endothelial cell loss at six months was 34.9% (±8.45). Mean visual acuity improved from 0.27 (±0.15) preoperatively to 0.70 (±0.32) at six months, with three patients reaching 0.8 decimal vision or better. Both techniques enhanced graft control in eyes with a deep anterior chamber. Use of an anterior chamber maintainer with a loose DMEK graft was particularly advantageous in an eye with an iris-claw IOL and silicone oil, facilitating manipulation and shortening unfolding time.
Conclusion: DMEK in vitrectomized eyes requires individualized surgical adaptation and appears safe, with outcomes comparable to non-vitrectomized eyes. Controlled elevation of anterior chamber pressure represents a useful adjunct, improving feasibility and safety in this complex patient group.

The RhoGEF Trio enhances the corneal endothelial barrier function

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Purpose: The corneal endothelium functions as a physiological barrier against fluids from the anterior eye chamber. A compromised barrier increases the risk of fluid accumulation in the corneal stroma, causing vision loss.

Currently, the only treatment is (partial) corneal transplantation. However, in 20%–40% of transplantations the endothelial graft (partially) detaches and requires re-intervention. Previous research has shown that overexpressing the N-terminus of the RhoGEF Trio (TrioN) leads to an increased cell surface area and barrier integrity in vascular endothelium. We have therefore studied the role of Trio in the corneal endothelium.
Methods: Corneal endothelial cells (CECs) were isolated from corneoscleral donor grafts that were unsuitable for transplantation. CECs were seeded in a cell density of 600 cells/mm^2 and grown to form a monolayer. CEC monolayers were then adenovirally transduced with a construct either for green fluorescent protein (GFP) or TrioN-GFP. Cells were fixed fifty hours post-transduction. Transduction efficacy and changes in cell-size were visualized by immunofluorescence imaging of GFP and filamentous actin respectively. Additionally, the corneal endothelial barrier function was measured using electrical cell-substrate impedance sensing (ECIS).
Results: CECs transduced with the TrioN-GFP construct showed a significant increase in the expression of filamentous actin, surface area and barrier function as compared to CECs transduced with GFP-only.
Conclusions: Our data show that TrioN successfully increases CEC surface area and junctional barrier function in corneal endothelial cells. These findings highlight the potential of the RhoGEF Trio as a strategy to restore the barrier integrity of damaged corneal endothelium.

Evaluation of novel open-source software for accessible vision simulations in Python

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Purpose: Optical simulations of the eye are commonly performed with commercial ray tracing software. The high costs of this software and required in-depth knowledge of optics hinders the use of these simulations in ophthalmic research. We therefore developed Visisipy, an open-source toolbox for vision simulations in Python. We evaluated Visisipy by comparing its output with three vision simulation studies.
Methods: Visisipy is available at https://demo.mreye.nl/visisipy. The software allows to create optical eye models using clinical measurements and perform ophthalmic analyses on these models. Results of three vision simulation studies were reproduced based on optical parameters specified by the authors. Central and peripheral Spherical Equivalent (SE) values were reproduced from a myopia study by Kneepkens et al. (2025); central and mid-peripheral Point Spread Functions from Van Vught et al. (2024); and ray tracing results from a fundus photography study by Haasjes et al. (2024). Furthermore, optical simulations were performed on personalized eye models for 27 subjects based on corneal topography, biometry, refractometry and MRI.
Results: All results from vision simulation literature were accurately reproduced. The optical simulations on 27 personalized eye models demonstrated that the software works for a wide range of ocular anatomies and the output is similar to that of commercial software like OpticStudio.
Conclusions: Visisipy provides a platform for accurate and reproducible vision simulations and makes vision simulations accessible for a wide audience of researchers and clinicians.

The natural history of myopia progression and visual acuity in children with Bornholm eye disease

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Introduction: Bornholm Eye Disease (BED) is an X-linked cone dysfunction syndrome characterized by reduced visual acuity (VA), high myopia, and color vision deficiency. This study evaluated the natural course of VA and refractive error in children with BED.
Patients and Methods: Twenty-five genetically confirmed BED patients were included in this retrospective cohort. Clinical data—VA, refraction, color vision, retinal imaging (optical coherence tomography (OCT), fundus autofluorescence (FAF))—and full-field electroretinography (ERG) were reviewed.
Results: At diagnosis, children (mean age 5.9 years) showed moderately reduced VA (mean 0.48 logMAR) and high myopia (mean –6.1 D). VA improved with age but remained subnormal (~0.2 logMAR in adolescence). Myopia progression was non-linear, with the fastest progression in early childhood; between ages 3 and 15, the mean increase was –3.0 D and largely independent of final refractive error. Color vision deficiency (predominantly protan) was present in 87.5%. OCT showed subtle central outer nuclear layer thinning, FAF revealed central hyperautofluorescence, and ERG demonstrated cone dysfunction with preserved rod responses. Pathogenic exon 3 splicing haplotypes were identified in 86%, with LIAVA–MVVVA and LVAVA–MVVVA commonly observed.
Conclusion: VA improves during childhood in BED but remains below normal. Myopia progresses slowly, and cone dysfunction is largely stable. Genotype–phenotype patterns may influence visual outcomes. BED is likely underdiagnosed and should be considered in boys with high myopia and unexplained VA reduction.

Results of intravitreal aflibercept injection and/or laser photocoagulation in the treatment of retinopathy of prematurity in a university hospital setting

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Purpose: Analysis of intravitreal aflibercept (IA) and laser photocoagulation (LT) as retinopathy of prematurity (ROP) treatment.
Methods: Retrospective observational cohort study of all preterms screened for ROP in the University Hospital of Antwerp between January 2023 and June 2025. Age, (re-)treatment need, timing, ROP regression, reactivation, zone and complications were studied; six months minimum follow-up.
Results: 141 preterms (282 eyes), median gestational age 27 weeks + 3 days (range 23w5d–32w4d), were screened. 103 eyes (36.5%) of 52 preterms developed ROP. Most severe ROP was recognized at median postmenstrual age (PMA) 34w5d. 25 eyes (24.2%) were treated. Thirteen eyes were treated with IA; regression was observed after median 8 days (range 4–14 days). Four eyes needed retreatment with LT after mean 63 ± 15 days. In nine eyes no adjuvant treatment was necessary. Twelve eyes were initially treated with LT; regression was observed after median 12 days (range 8–25 days). Adjuvant IA was applied after 21 days in two eyes. First ROP treatment at mean PMA of 36w4d (±22.1 days) for IA and 37w3d (±13.8 days) for LT. IA was predominantly used for ROP in zone I and posterior zone II (84.6%) and LT for zone II (83.3%). Complications: IA none. LT two; self-clearing vitreous hemorrhage, mild macular dragging.
Conclusions: Aflibercept might be a relevant monotherapy for zone I and posterior zone II ROP, without the need for adjuvant treatment in the first months and might induce ROP regression earlier than LT. Laser therapy as initial treatment is relevant in older preterms with peripheral ROP.

Joint ophthalmological care for children with orbital vascular malformations

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Introduction: Orbital vascular malformations in infants present unique diagnostic and therapeutic challenges due to their rarity and potential for vision-threatening complications. We report an intriguing case of a neonate with an infraorbital vascular malformation with extreme exophthalmia and direct threat on the optic nerve and eye. This case demonstrates how a multidisciplinary approach within our vascular anomalies expertise center enables successful management of complex patients.
Methods: A healthy one-month-old boy was referred for severe unilateral subluxation and progressive proptosis, exacerbated during pressure-increasing episodes. Examination revealed complete luxation of the left eye with intraocular pressure rising from 8 to 81 mmHg, alongside partial optic nerve excavation. Initial differential diagnoses of proptosis in neonates included congenital, infectious, inflammatory, neoplastic, and vascular causes; the clinical presentation strongly suggested a vascular origin.
Results: MRI and CT imaging confirmed an intraorbital mass. Additional prone-position MRI mimicking a Valsalva maneuver demonstrated features consistent with a venous malformation, classified as a slow-flow vascular anomaly present at birth. Management required a multidisciplinary approach involving ophthalmology, pediatric hematology, dermatology, radiology, and interventional radiology. Treatment combined systemic sirolimus with local sclerotherapy using bleomycin foam. After nine months of sirolimus and three sclerotherapy sessions, the patient exhibited marked improvement without pressure-induced proptosis.
Conclusion: This case highlights the importance of early recognition, advanced imaging techniques, and collaborative care in managing complex orbital vascular malformations. Targeted interventions offer promising outcomes while preserving ocular function.

Placental hypoxia as a risk factor for ROP

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Background: This study investigates the contribution of placental inflammation and uteroplacental malperfusion in the development of retinopathy of prematurity (ROP) with the aim of refining the risk profile for (severe) ROP and enhancing targeted screening of high-risk neonates.
Methods: This retrospective cohort study included 91 preterm neonates (born <30 weeks gestational age (GA) with body weight (BW) <1250 g or born 30–32 weeks of GA and/or BW 1250–1500 g with an established risk factor) born at or transferred to the Neonatal Intensive Care Unit at Leiden University Medical Centre (2018–2022). Maternal and neonatal clinical data were extracted from health records, and placental specimens were re-examined histopathologically according to the Amsterdam Placental Workshop Group Consensus Statement. The primary outcome was the development of ROP, including severe forms. Multivariate regression analyses were performed to adjust for confounding variables.

Results: Fetal hypoxia was significantly associated with any-stage ROP (OR = 6.8; 95% CI, 1.6–29.5) and severe ROP (OR = 8.9; 95% CI, 1.002–78.7). No significant associations were identified between placental inflammation or uteroplacental malperfusion and ROP.
Discussion: Fetal hypoxia emerged as independent risk factor for both ROP and severe ROP, emphasizing the interaction between prenatal and postnatal determinants of disease. These findings suggest that intrauterine conditions play a contributory role in the pathogenesis of ROP, underscoring the importance of early identification of prenatal risk factors. Early recognition may facilitate targeted screening and interventions, potentially reducing the incidence of ROP and improving neonatal outcomes.

Accessibility of refractive corrections in Dutch children

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Introduction: Approximately 635,000 children under 18 (19.1%) in the Netherlands are affected by refractive errors. There have been growing concerns among professional medical associations, advocacy groups, and charitable organizations about the accessibility of glasses for children, as 10.1% live under or slightly above the poverty line. Our aim was to investigate the accessibility of refractive corrections, which are crucial for children's (eye) health and development.
Methods: A digital survey was distributed among 2106 parents/caretakers of 3297 children aged 0–17 years. Questions focused on the nature of refractive errors and corrections, and barriers to accessibility.
Results: Of all children, 581 (17.6%) had glasses and/or contact lenses for myopia (<−0.5D; 57%) or hyperopia (>+0.5D; 31%, others unknown). The median expenses were €250, - in the past 12 months. Expenses were higher with increasing refractive error up to €363,- in case of <−6D or >+6D. Basic insurance was accessed by 7%, 79% did not meet requirements, and 5% received support from family, municipalities or charities. Moreover, 18% postponed the purchase, chose a cheaper option, or did not replace broken or outdated prescriptions; 7% needed to make budget decisions at the expense of essentials, including groceries, clothing, travel, investments, e.g. home improvements. In 16.5% of children without glasses and assumed vision difficulties ($n = 103$), parents avoided screening due to potential high costs.
Conclusion: The accessibility of refractive corrections for children seems limited by affordability concerns. We recommend that policymakers develop strategies to reduce health inequalities and ensure equal opportunities by making refractive corrections available to all children.

Predictive value of choroidal thickness for the incidence and progression of myopia

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Purpose: To assess whether choroidal thickness (ChT) predicts long-term myopia incidence and progression in a population-based children's cohort.
Methods: Children underwent cycloplegic refraction, axial length (AL) measurements, and OCT-imaging at ages 9, 13 and 17 years. The choroid was automatically segmented and thicknesses were calculated using AI. ChT was compared between myopes and non-myopes. Stepwise regression models evaluated the predictive value of ChT and AL for myopia incidence and progression.
Results: At ages 9, 13, and 17, 1175, 3585, and 3607 children were included. Myopia prevalence rose from 13% to 35%. From 9 to 17 years, mean ChT increased from 257 to 296 μm in myopes and from 293 to 346 μm in non-myopes, remaining consistently lower in myopes ($p < 0.001$). Higher AL at age 9 predicted incident myopia (HR: 1.98; CI: 1.51–2.60), unaffected by adding ChT (HR: 1.93; CI: 1.47–2.55). ChT did not predict myopia onset before or after AL adjustment. C-statistics were 0.63 for ChT and 0.72 for AL, with no improvement when combined. At age 13, greater ChT predicted less axial length growth up to age 17 (β −0.004 mm per year per 100 μm; CI: 0.000–0.008), independent of baseline AL. Longer baseline AL consistently predicted faster progression.
Conclusions: ChT at age 9 did not predict myopia onset. ChT at age 13 was associated with subsequent AL growth, independent of baseline AL. Although the choroid may play a causal role in myopia development, its value for long-term prediction of myopia incidence and progression appears limited.

Confounders in the early glasses study including rapid prescription of glasses for squint

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Introduction: The Early Glasses Study assesses the relation between refractive error and odds to develop amblyopia or esotropia, and whether glasses can prevent their development. We assessed confounders.
Patients and Methods: At Entry Exam 601 healthy children aged 14.5±1.7 months received orthoptic examination followed by retinoscopy in cycloplegia. Children with refractive error >AAPOS Criteria 2003 were randomized into glasses (N=28) or not (N=24) and all 601 were followed-up until age 4. In children whom were prescribed glasses after detection of esotropia or amblyopia, at Entry Exam or during follow-up, the influence was studied of prompt or delayed prescription of glasses on outcome.
Results: 1.7% of parents reported low level of education (13% in general population). In the second year of life refraction changed between −2.375D and +2.75D in randomized children. Among 12 whom were prescribed glasses after squint onset, esotropia was cured and amblyopia did not develop in six with a median interval between squint onset and glasses of 0.5 months. In six others esotropia persisted or amblyopia developed with a 1.9 months interval. Study children got glasses faster as compared to siblings with similar disease and their esotropia was more often cured and amblyopia developed less often.
Conclusion: As refraction changes considerably in the second year of life, a single measurement is insufficient. The influence of prompt prescription of glasses after squint onset on outcome was large. Hence, we advise children aged 1–3 suspected of squint to be examined within a week and prescribed glasses within two weeks.

Early prediction of severe retinopathy of prematurity using clinical risk factors and continuous physiological data

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Purpose: To develop a prediction model for severe retinopathy of prematurity (ROP) by integrating clinical risk factors and continuous physiological data collected during phase 1 of ROP development.
Methods: Preterm infants screened for ROP at Erasmus University Medical Center Rotterdam between 2018 and 2023 were eligible. Severe ROP was defined as type 1 or 2 ROP, treated ROP, or ROP with pre-plus disease. Clinical and physiological data from birth to 32 weeks post-menstrual age were collected and summarized within 12-h timeframes. Machine learning classification models were developed, and evaluated with five-fold cross-validation. Performance metrics include area under the receiver operating curve (AUROC), sensitivity and specificity.
Results: Of 349 screened infants, 72% (n=170) had any stage ROP and 23% (n=80) had severe ROP. After removing highly correlated variables, physiological data alone yielded a maximum AUROC of 0.66, using mean fraction inspired oxygen and oxygen saturation. Adding clinical data resulted in a mean cross-validated AUROC of 0.78, 74% sensitivity and 72% specificity. Top predictors included gestational age, small for gestational age, several neonatal morbidities (intraventricular hemorrhage, necrotizing enterocolitis) and treatments (transfusions, steroids, inotropics, mechanical ventilation), maternal factors (hypertension, diabetes, fever, steroids and preterm pre-labor rupture of membranes), and skewness of heart rate and perfusion index.
Conclusions: Continuous physiological data showed moderate predictive value for severe ROP, with model performance improving when clinical data was added. Adjusting data summarization methods for continuous data might improve performance. Further evaluation of the data should also explore time-to-event modeling.

Expanding the phenotypic and genetic spectrum of oculocutaneous albinism type 8

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Purpose: To describe the phenotypic spectrum of the recently discovered oculocutaneous albinism type 8, caused by mutations in the *DCT* gene.
Methods: We compared the pigmentation of skin and hair of eight patients carrying two variants in the *DCT* gene from France and the Netherlands to that of their siblings and/or parents. We collected genetic, ophthalmic, and electrophysiological data and supplemented these with

the data of all six previously described patients in the literature. We used grading schemes for foveal hypoplasia (0–4), with grade 0 referring to no abnormalities and grade 4 indicating the absence of all foveal structures.
Results: All patients had only mild skin and hair hypopigmentation. Visual acuity (VA) ranged from 0.3–1.0 Snellen, 12/14 had nystagmus, 8/14 had iris transillumination, fundus pigmentation was normal to mildly reduced, grade 0–2 foveal hypoplasia, and misrouting was identified in all patients evaluated (5/5).
Conclusions: Pigmentation of skin and hair is only mildly affected in patients with oculocutaneous albinism type 8. The ocular phenotype is also mild, with VA at the better end of the range seen in albinism, and mild foveal hypoplasia. The *DCT* gene encodes DOPAchrome tautomerase (DCT), an enzyme required for the conversion of dopachrome into dihydroxyindole carboxylic acid and eventually the synthesis of black melanin. In the absence of DCT, dopachrome is converted into dihydroxyindole, resulting in the synthesis of brown melanin instead. The intact brown melanin synthesis pathway is likely responsible for the mild phenotype.

Global axial length centile charts: A comprehensive study across continents in the CREAM-KIDS consortium

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Purpose: Myopia is increasingly common worldwide and poses significant public health challenges due to the pathological complications of axial elongation. Axial length (AL) is the primary target for myopia management, creating a need for population-specific reference AL charts. This study aimed to generate region- and sex-specific reference centile charts for AL.
Methods: We meta-analysed 559,938 individual AL measurements from 147,573 children. Most participants were from East Asia (84%, age 6–18 years), followed by Europe (12%, age 6–21 years) and Australia (4%, age 6–21 years). After assessing comparability, European and Australian data were combined. Centile estimates were extracted from multi-Gaussian models fit to age- and sex-specific AL distributions and smoothed using weighted cubic splines optimised by the Generalized Akaike Information Criterion.
Results: AL distributions differed by sex and region: in females, median AL at age 7 years was 22.35 mm in Europeans/Australians and 22.67 mm in East Asian cohorts; by age 18, these were 23.31 mm and 24·36 mm, respectively. Regional differences were more pronounced at higher centiles. Modelled centile curves showed excellent agreement with empirical values (Pearson r = 0.99) with minimal bias (–0·08 mm to +0·02 mm).
Conclusions: These centile charts provide region- and sex-specific AL references for childhood and adolescence, offering a valuable tool for detecting abnormal eye growth and supporting targeted myopia prevention and management strategies.

Long term visual outcome and complication rate in paediatric cataract surgery

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Purpose: Evaluation of long term visual outcome and postoperative complications of paediatric cataract surgery.
Methods: Retrospective, single centre cohort study presenting data of children <16 years, operated for paediatric cataract between May 2011 and January 2025 with >1 year follow up. Etiology, age at detection and surgery, non-surgical treatment preceding surgery, visual outcome and complications will be reported. Patients were divided in 4 categories: unilateral congenital (CU) or acquired (AU), bilateral congenital (CB) or acquired (AB).
Results: A total of 222 eyes of 155 patients were included. Unilateral cataract was present in 52.9%, bilateral cataract in 47.1%. Median age at diagnosis: CU 12.5 (0.5–176), AU 92.5 (90.7–192), CB 3.0 (0.5–165) and AB 72 (4.5–174) months (range). Median age at surgery, performed by 1 surgeon (S-D): CU 29.5 (1.0–177), AU 105.1 (98.4–193), CB 7.0 (1.0–167) and AB 92 (5.0–189.0) months (range). Median follow up was 6 (1–15) years. Afakia in 75, pseudofakia in 147 eyes. Before surgery 42 children were treated with patching (10) or patching with homatropine 2% (32). Median visual acuity at start and follow up was CU 0.2 (0.01–0.5)/0.4 (0.01–1.2), gain 0.5 (0.02–1.0); AU 0.1 (0.15–0.40)/0.8 (0.7–1.2), gain 0.6 (0.5–1.1); CB 0.3 (0.08–0.4)/0.7 (0.02–1.50), gain 0.6 (0.4–1.10) and AB 0.3 (0.01–0.8)/1.0 (0.1–1.5), gain 0.8 (0.05–1.3). During follow up 9 eyes needed surgery for membrane formation (2CU), secondary glaucoma (2CU), wound leakage (1AB) and recurrent bleeding from PFV vessels in posterior pole (3CU).
Conclusions: In this cohort, visual outcome was comparable to other studies but complication rate was low.

Iso-accommodation correlates with stereopsis in orthoptic patients with treated amblyopia, strabismus or refractive error

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Purpose: In the Early Glasses Study, 5/601 children had anisometropia >1.5D at 14.5 ± 1.7 months (10.1007/s00417-024-06621-8). In 3 of these, anisometropia resolved, disqualifying anisometropia as amblyopia risk factor at age 1, and suggesting that amblyopia may cause anisometropia instead, as previously shown in monkeys (10.1016/0042-6989(94)00239-i). Because emmetropization is a local retinal process (10.1016/j.visres.2024.108402), insufficient accommodation could impede emmetropization of the amblyopic eye by defocus.
Methods: Orthoptic patients aged 7–12 wearing glasses prescribed after retinoscopy in cycloplegia were recruited (10.1111/aos.15016). Bilateral refraction was measured continuously at 50 Hz using the PowerRefractor II. An Arial 9pt reading text, to be read aloud, was approached manually from 1 m to 10 cm taking approximately 10 s. Iso-accommodation (IsoAccom) was defined as a difference in refraction not exceeding 1D.
Results: Of recordings in 204 children, 145 included measurements from both eyes. IsoAccom was not related to age. IsoAcc was found in orthoptic patients with worst VA 0.2logMAR, +5.25D spherical equivalent, 4.25D astigmatism and 2.5D anisometropia, but all of these had Lang 200″ or TNO 60″ stereopsis. Although most with 200″ stereopsis had IsoAccom, a few with 60″ did not.
Conclusions: IsoAcc correlated strongly with stereopsis. Children with high hypermetropia, astigmatism or anisometropia had IsoAccom only when stereopsis was Lang 200″ or TNO 60″. Three IsoAccom cases with anisometropia 1.6-2.5D had TNO 60″, only possible with consistent wearing of glasses from young age. In amblyopia with lack of stereopsis, defocus of the retinal image of the amblyopic eye by insufficient accommodation could impede emmetropisation of that eye and thereby causes anisometropia.

Wide field of view measurement with multi-camera stereo eye-tracking

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Purpose: Conventional remote eye-trackers are limited to a horizontal field of view (FOV) of approximately 30°, restricting their use in applications requiring large gaze angles, such as strabismus assessment and traffic-related gaze analysis. This study aims to develop and validate a stereoscopic eye-tracking system with an extended FOV of 100°.
Methods: An experimental setup was constructed with five cameras, four infrared light sources, and three monitors. Dedicated algorithms were developed for robust assignment of detected light source reflections to their corresponding light sources and for dynamic camera selection, ensuring optimal image quality and reliable tracking across the entire FOV. Gaze reconstruction was performed offline using multi-view triangulation and a single-point calibration for the optical-visual axis deviation. Data were collected from healthy adult participants ($n = 5$), and spatial accuracy was quantified across the FOV.

Results: The system demonstrated high spatial accuracy (mean absolute error <1°±0.3°) for fixations up to 22° eccentricity and good spatial accuracy (<1.2°±0.9°) for targets up to 40° eccentricity. For targets beyond 40° eccentricity, fixation was often not achieved.
Conclusions: A stereoscopic eye-tracking system with a horizontal FOV of up to 80° is feasible and provides accurate gaze measurements with minimal calibration effort. The system is particularly suitable for clinical and research applications requiring large gaze excursions, such as strabismus assessment and traffic studies. Further optimization of the eye model and hardware may improve accuracy at higher eccentricities.

Effect of preoperative botulinum toxin injection two weeks before vertical rectus transposition in complete sixth nerve palsy

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Purpose: We evaluated the effect of preoperative botulinum toxin type A (BTXA) into the antagonist muscle in patients with complete 6th nerve palsy, two weeks before the Hummelsheim procedure, to release contracture.
Methods: All patients with an acquired complete 6th nerve palsy were eligible. BTXA was injected into the ipsilateral rectus muscle two weeks before Hummelsheim transposition. Ocular deviations were assessed by alternating prism cover test at distance preoperatively, 2 weeks post-BTXA, 3 and 12 months post-Hummelsheim. Outcomes were angle reduction, diplopia in primary position, and postoperative stability. Effect size was analysed with linear regression including age, preoperative angle, etiology, and abduction limitation as predictors.
Results: 31 patients (51±20 years) identified. Etiologies included tumor (15), head trauma (11), and other causes (5). Mean angle of deviation was 51±10PD preoperatively. Two weeks after the BTX angles were 18±21PD; three months post-Hummelsheim there was an exotropia 5±16PD. After one year mean angle was 1±10PD; six remained overcorrected. Overall, mean effect was a reduction in esotropia of 48±19PD. The only significant predictor was preoperative angle of strabismus (B=0.7; p=0.003). Twenty-four (77%) had no diplopia in primary position, three needed additional surgery.
Conclusion: Preoperative BTXA two weeks before Hummelsheim procedure enhanced outcomes, yielding a 48PD reduction and stable alignment after one year. Most patients (77%) were diplopia-free in primary position supporting the use of BTXA prior to vertical rectus transposition in complete 6th nerve palsy.

Visual field interpretation in pituitary adenomas: How reliable are ophthalmology residents and optometrists?

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Introduction: Nonfunctioning pituitary adenomas (NFAs) frequently cause visual field defects (VFDs) due to compression of the optic chiasm. Accurate interpretation of visual field tests is essential for diagnosis, surgical decision-making, and postoperative monitoring. Although ophthalmology residents and optometrists routinely interpret visual fields in clinical practice, little is known about the reliability of their assessments compared with expert neuro-ophthalmologists. This study evaluates the accuracy of pre- and postoperative VFD interpretation by residents and optometrists in patients with NFAs.
Methods: In this cross-sectional observer performance study, 81 Humphrey Field Analyzer (30-2 SITA Fast) tests from patients who underwent transsphenoidal surgery (2005–2016) were independently assessed by eight ophthalmology residents and two optometrists. Each assessment included determining (1) the presence of a VFD, (2) the VFD pattern, and (3) postoperative change. Ratings were compared with expert consensus from two neuro-ophthalmologists. Cohen's kappa was used to evaluate agreement, with sub-analyses excluding cases marked as "not assessable" or "supervision required."
Results: Agreement for identifying abnormal visual fields was substantial to almost perfect (κ=0.600–0.856), improving further when non-assessable cases were excluded (κ=0.639–0.891). Pattern recognition showed moderate to substantial agreement (κ=0.467–0.702). Assessment of postoperative change demonstrated moderate to substantial agreement (κ=0.446–0.756), with higher values when uncertain cases were excluded. No consistent improvement across residency years was observed; optometrists performed comparably to residents. Expert individual assessments showed near-perfect agreement with their final consensus.
Conclusion: Ophthalmology residents and optometrists demonstrated substantial reliability in identifying VFDs and postoperative changes in patients with NFAs. While pattern recognition was less consistent, overall findings suggest that both groups can meaningfully contribute to clinical decision-making, provided that uncertainty is acknowledged and supervision is sought when needed.

Normative modelling of white-matter pathways in the visual system

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Purpose: Classical case control research relies on group-level differences, averaging out individual variation and making findings less informative on an $N=1$ basis. We aim to create normative ranges for the visual system's major white matter tracts to allow assessment of an individual relative to the population wide norm.
Methods: We used the population-based UKBioBank dataset (project number 23668 and 705478). After data-driven MRI quality control, we selected over 16000 individuals (age range: 45–81) without any known ophthalmologic or neurologic disease. We analysed three white matter bundles, optic tracts, and optic radiations conveying central (<3 degrees) and peripheral (>15 degrees) visual field information. These white matter tracts were used to extract advanced (microstructural) parameters, after which these were modelled against age using Generalized Additive Models of Location, Shape and Scale, as used by the WHO for the construction of growth charts. We evaluated the sample size sufficiency for estimation of different centiles of the constructed normative models.
Results: Normative models for four white-matter integrity measures, modelled across age, were constructed for the optic tract and the central and peripheral optic radiations. Used sample sizes produced models which are sufficient in robustly estimating the 5th up to the 95th centile values.
Conclusions: The created normative models enable quantification of the deviation from the norm on an individual (patient) level, in the white matter pathways that transmit retinal signal to the visual cortex. These can be used as predictive markers for treatments aiming to restore vision at the retinal level.

Retinal vascular features to predict open-angle glaucoma

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Purpose: Next to intraocular pressure (IOP)-related mechanisms, vascular pathways have been hypothesized in the mechanisms leading to open-angle glaucoma (OAG). We aimed to assess the relevance of retinal vascular alterations for OAG prevalence and incidence using a fully automated deep-learning pipeline (VascX).
Methods: This study was conducted within the Rotterdam Study, a prospective population-based cohort study. All participants underwent ophthalmic examinations at all visits, including IOP measurements, fundus imaging, and perimetry. Color fundus images were processed using VascX to extract macular vascular features. Multivariable logistic regression (for cross-sectional data) and Cox proportional hazard regression (for longitudinal data) were performed, and stratified by OAG polygenic risk score (PRS) tertiles.
Results: A total of 13,099 participants were included (including 320 prevalent OAG and 176 incident OAG-cases). Cross-sectionally, larger arterial Central Retinal Equivalent and higher arterial vessel density were associated with a lower odds of prevalent OAG (OR: 0.70, 95% CI: 0.59–0.85 and OR: 0.65, 95% CI: 0.53–0.79, respectively). OAG cases showed significantly larger declines over time in these vascular features compared to non-OAG cases. Longitudinally, higher arterial vessel density reduced the risk of incident OAG (HR: 0.74, 95% CI: 0.58–0.95). A predictive model incorporating arterial vessel diameter, clinical factors, and PRS achieved high discrimination for both prevalent (AUC 0.80) and incident OAG (C-statistic 0.82).
Conclusion: Automated profiling of macular retinal vasculature using the VascX pipeline identifies structural biomarkers (particularly arterial caliber and density) associated with OAG. Integrating these objective vascular features may enhance the predictive performance of OAG models.

One year outcomes of combined phacoemulsification with excimer laser trabeculostomy in patients with glaucoma

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Introduction: To investigate the effect of Excimer Laser Trabeculostomy (ELT) as a surgical treatment for patients with glaucoma.

Patients and Methods: Patients with both cataract and glaucoma underwent combined phacoemulsification with lens implantation and ELT (Elios Vision Inc., Los Angeles, CA, USA). Data was included for analysis if a minimum follow-up period of 1 year was available.
Results: A total of 59 eyes from 37 patients were treated. The mean preoperative intraocular pressure (IOP) was 16.2 ± 5.0 mmHg. Postoperative IOP was reduced to 13.8 ± 3.9 mmHg at 1 month, 12.9 ± 3.1 mmHg at 3 months, 13.2 ± 3.8 mmHg at 6 months, and 13.0 ± 3.0 mmHg at 1 year. The mean number of glaucoma medications decreased from 2.4 ± 1.3 preoperatively to 1.8 ± 1.2 at 1 month, 1.4 ± 1.2 at 3 months, 1.5 ± 1.3 at 6 months, and 1.7 ± 1.2 mmHg at 1 year. Both IOP and the number of medications were significantly lower at all time points from 1 month postoperatively compared to preoperative levels ($p < 0.001$). Reversible microhyphema was observed in 7 eyes, and transient IOP spikes occurred in 12 eyes (7 eyes >10 mmHg, 5 eyes 5–10 mmHg) in the early postoperative period. No other complications were noted during the follow-up period.
Conclusion: In this patient cohort, ELT appeared to be a safe treatment option for patients with cataract and glaucoma. Further research and long-term follow-up are needed to evaluate the long-term effects of this treatment.

Screening for glaucomatous visual function loss with SONDA-based eye movement perimetry

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Purpose: The gold standard for visual field assessment is standard automated perimetry (SAP). SAP can be too demanding for some individuals. Continuous visual stimulus tracking (SONDA: Standardized Oculomotor and Neurological Disorders Assessment) simplifies the task to following a moving stimulus on a screen. Here, we evaluated the screening performance of SONDA-based eye movement perimetry (SONDA-EMP) in glaucoma using two implementations: the experimental setup (SONDA-Eyelink) and the clinic-ready version (SONDA-Neon).
Methods: SONDA-Eyelink and SONDA-Neon measurements were performed in 100 cases with glaucoma (36 early, 36 moderate, 28 severe glaucoma) and 100 age-similar controls. Participants monocularly tracked a moving stimulus (Goldmann size-III) at 40% contrast (both setups) and 160% (SONDA-Eyelink). Eye movements were continuously recorded. Outcome was the agreement between gaze and stimulus position. We used previously collected glaucoma case-control data to build a continuous 'glaucoma screening score'. This score was used for an ROC-analysis applied to the current, independently collected dataset. We predefined good screening performance as: at 95% specificity, a sensitivity of at least 50%, 90%, and 100% for early, moderate, and severe glaucoma, respectively.
Results: At 95% specificity, the sensitivity of SONDA-Eyelink was 58, 94, and 100% at 40% contrast and 56, 97, and 100% at 160% contrast for early, moderate, and severe glaucoma, respectively. Sensitivity was 53, 94, and 100% for SONDA-Neon. Cases rated SONDA-EMP consistently easier and less tiring compared to SAP.
Conclusions: SONDA-EMP offers a novel and intuitive approach that meets the need for fast and accessible methods to screen for visual function loss in glaucoma.

Follow-up performance and misclassifications of a deep-learning algorithm for glaucoma detection in the Rotterdam Study

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Introduction: We validated a deep-learning (DL)-algorithm for detecting perimetric glaucoma in a population-based cohort. We evaluated longitudinal performance, misclassification patterns, and differences between macula- and disc-centered color fundus photographs (CFPs).
Patients and Methods: Data were obtained from the prospective population-based Rotterdam Study. Cross-sectional performance was assessed by sensitivity at 95% specificity and area under the receiver-operating-characteristic curve (AUC). Longitudinal data were evaluated for all perimetric glaucoma cases. False negatives (FNs) and the 100 false positives (FPs) with the highest glaucoma-likelihood scores were qualitatively reviewed. Performance differences between macula- and disc-centered CFPs were determined in a paired subset.
Results: Of all CFPs, 4076 (3.9%) were ungradable. After selecting one follow-up moment, 19088 baseline CFPs were included (ages 50–80 years). Sensitivity at 95% specificity was 70.6% (AUC 0.911). True positives (TPs) showed more advanced visual field loss than FNs (median MD −10.8 vs −5.8 dB; $p < 0.0001$). Among FNs with follow-up ($n = 24$), 9 were detected later by the

algorithm (median 5.9 years). Of TPs with earlier CFPs available ($n=61$), 33 were identified before conversion to perimetric glaucoma (median lead-time 6.4 years). FNs frequently exhibited image-quality issues. All 100 evaluated FPs displayed structural glaucomatous features. Disc-centered CFPs outperformed macula-centered CFPs (AUC: 0.940 vs 0.905; $p=0.022$).

Conclusion: The DL-algorithm primarily detected more advanced glaucoma, with delayed detection in a subset of FNs. Misclassifications were related to image quality or structural glaucomatous findings. ONH capture is essential for reliable glaucoma assessment.

Promoting psychological well-being in younger adults with visual impairment using a stepped care model

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Purpose: Depression and anxiety affect approximately one in three people with visual impairment, underscoring the need for psychosocial support. While a stepped care model is already available for adults over 50, younger adults experience distinct challenges and support needs. This study examined these specific issues and adapted the stepped care model, including an online self-help course for adults under 50.

Methods: A concept mapping study explored the main psychosocial needs and everyday challenges faced by younger adults with visual impairment. The study involved 52 individuals with visual impairment and 27 health care providers. The findings guided refinements to both the stepped care structure and the online course content, ensuring greater age relevance. Usability testing with eight participants with visual impairment then assessed the accessibility and engagement of the online course. Based on this feedback, several improvements were made to enhance the course's accessibility.

Results: The adapted stepped-care model is currently being implemented at Amsterdam UMC and Royal Dutch Visio with 40 participants. It consists of four steps: (1) watchful waiting, (2) online self-help, (3) problem-solving treatment, and (4) referral to psychological care. The online course includes modules on acceptance, relaxation, fatigue management, communication, resilience, and relapse prevention. Participants complete the modules from the comfort of their homes, receive guidance from a social worker by phone, and engage with diaries, exercises, and peer stories.

Conclusion: Adapting and implementing a stepped-care model for younger adults with visual impairment is both feasible and clinically relevant, possibly promoting mental well-being and enhancing self-management.

Relationships between vision, hearing, and daily life challenges

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Purpose: Dual sensory loss (DSL), in which impairments in vision (VI) and hearing (HI) are combined, significantly affects daily life, for example with respect to information access, mobility, communication, and fatigue. The aim of this study was to investigate whether people with DSL experience more difficulties on these domains than individuals with single or no sensory impairment.

Methods: 179 participants (54% female; mean age 58) took part in this cross-sectional study ($n=48$ DSL, $n=39$ VI, $n=32$ HI, and $n=60$ normal vision and hearing (NVH)). Daily life challenges were assessed using validated questionnaires. Visual acuity, visual field, contrast sensitivity, hearing threshold, speech understanding, and directional hearing were also measured. Differences between groups were analyzed using (M)ANOVA.

Results: Preliminary analyses show significant group effects on all domains ($p<0.001$). With respect to information access, mobility, and fatigue, post hoc analyses revealed that the DSL group scored significantly worse than the HI and NVH groups ($p<0.002$). Regarding communication, the DSL group had significantly worse scores than the VI and NVH groups ($p<0.001$).

Conclusions: It appears that in certain life domains, individuals with DSL exhibit functioning similar to those with VI, while in other areas, it more closely resembles that of HI. Future analyses may uncover that (interactions between) visual and auditory functions are associated with enhanced or limited functioning on these domains. The findings of this study emphasize the clinical importance of taking both vision and hearing into account, as this can guide more tailored healthcare for people with DSL.

Stakeholder views on the current age-related macular degeneration care path: A qualitative study

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Introduction: Although treatment options to preserve vision in exudative age-related macular degeneration (AMD) have expanded significantly in recent years, the pressure to deliver high-quality care in overcrowded eye clinics has become an increasingly urgent challenge. The aim was to investigate barriers and facilitators within the current AMD care path.
Methods: We conducted semi-structured interviews with 15 healthcare professionals, 15 patients, and 15 significant others. The interviews explored the patient journey, eliciting lived experiences, stakeholder perspectives on the AMD care path, perceived barriers and facilitators, and communication needs regarding the organization and delivery of care.
Results: For patients themes emerged regarding the diagnostic experience, clarity of information, and the psychosocial burden of AMD, particularly uncertainty about disease progression. Fragmented information left patients unsure about expectations and next steps. Healthcare professionals reported structural constraints, particularly limited consultation time, which hindered thorough explanations and discussions on daily-life implications. Themes for significant others included the need for proactive support to prevent caregiver overload and for information about the care path.
Conclusions: Stakeholders experience multiple challenges within the current AMD care path, especially related to diagnosis, information provision, consultation time, and psychosocial impact. Professionals further highlighted systemic pressures that restrict their capacity to offer comprehensive support. Significant others indicated additional support needs within the care path. These insights will inform the forthcoming co-creation phase and guide the development of a more patient-centred and coherent care path.

The consolation of the visually impaired Helen Keller as extracted from her diary manuscript 1936–1937, an explorative case study

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Purpose: "La médecine c'est guérir parfois, soulager souvent, consoler toujours" is a centuries old aphorism with apparent value since its widespread use, however, in the current ophthalmologic literature there is few academic attention for consolation. We therefore intend to explore on experienced consolations and started by close-reading of a diary of a single blind person in a specific and well-known historic setting.
Methods: This case study reviews the diary notes 1936–1937 of the visually impaired Keller starting 15 days after Keller's life companion passed away. We applied a strategy known as enactive understanding of emotions and experience: Experienced consolatory interactions (ECI's) were defined as the written encounters by Keller between her and her socio-material environment with a positive transformative effect on her experience. The different ECI's were identified and scored for the setting (9 categories), the type of distress (5 categories), and the used consoling strategy (8 categories).
Results: 167 ECI's were identified. In the first month, the less nuanced distresses “unspecified” and “feeling blue” made 67.7% out of the ECI's versus the 32.3% more nuanced distresses “struggling the environment”, “grieving loss”, and “worrying”. In the following 5 months, the less nuanced made up 46.9% versus 53.1% more nuanced ($p<0.001$).
Conclusion: The authors suggest that the emotional granularity of Hellen Keller increased in the half year after the loss of her life companion. This study offers a model of studying consolation to facilitate future investigations, eventually leading to insights that will improve quality of life of the visually impaired.

Evaluation of artificial iris implantation in traumatic iris injury cases

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Introduction: Traumatic iris injuries can result in significant photophobia (light sensitivity) and cosmetic deformities due to iris defects. Artificial iris implantation offers a solution to restore both the function and appearance of the iris in such cases. This report evaluates six consecutive cases of artificial iris implantation in eyes with traumatic iris injuries or related iris defects, focusing on outcomes regarding symptom relief and cosmetic improvement.
Methods: Six cases were included in this evaluation, four eyes with post-traumatic iris damage, one eye with iatrogenic mydriasis, and one eye with congenital aniridia. One eye underwent implantation of a prosthetic iris (Reper) in the ciliary sulcus. Three eyes received a fiber-free artificial iris (HumanOptics) implanted in the ciliary sulcus. One eye received a fiber-free artificial iris implanted in the capsular bag. The sixth case involved a patient with congenital aniridia and aphakia, who underwent scleral fixation of a fiber-containing artificial iris. All procedures were performed with minimal surgical intervention and served as secondary treatments to improve both photophobia and cosmetic appearance.
Results: All surgeries were completed without complications, and the postoperative outcomes were excellent. Photophobia symptoms were completely resolved in all six cases. One eye developed pigment dispersion glaucoma due to iris chafing, which was successfully managed with extra peripheral YAG laser iridotomy in recipient and anti-glaucoma medication.

Effectively informing children and parents in the evolving practice of myopia management

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Purpose: To investigate and improve patient education in myopia management by developing practical information tools for children, parents, and professionals.
Methods: The study comprised a design phase with online surveys among 47 Dutch parents of children with myopia and 90 orthoptists, with response rates of 24% (47/200) and 23% (90/400), respectively. Findings informed focus groups to identify needs for information tools, guiding the co-development of a practical decision-aid in the implementation phase. The tool was then evaluated in clinical practice and refined based on feedback.
Results: Nineteen of 47 parents were satisfied with information provided to their child; among them 15/19 were informed using visual aids or analogies versus 4/24 of the less satisfied parents ($p<0.001$; 4 n.a.). Thirty of 47 parents were satisfied with information provided to themselves; among them 24/30 felt fully involved in decision-making about treatment versus 6/15 of less satisfied parents ($p=0.008$; 2 n.a.). For having sufficient time to decide treatment, this was 18/22 versus 4/11 ($p=0.009$; 14 n.a.). Orthoptists differed in treatments regularly recommended: low-/high-dose atropine (74/90; 14/90), inhibiting glasses/soft contactlenses (56/90; 30/90), and ortho-keratology (10/90). Seventy-three of 90 orthoptist expressed the need for additional information tools. Based on these findings, decision-aid cards were developed and clinically tested; 33/36 parents and 15/20 orthoptists reported satisfaction. After final adjustments, the tool comprised five cards addressing children, parents, schools, treatment options, and lifestyle recommendations.
Conclusions: Decision-aid cards incorporating diverse perspectives on myopia management can streamline consultations, strengthen shared decision-making, improve patient satisfaction, and improve intervention outcomes.

Shifting patterns of visual impairment and blindness: Prevalence and cause- and age-specific trends in 30 years of population-based data

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Purpose: Demographic changes, including population ageing and rising myopia prevalence, may have altered the prevalence and causes of blindness and visual impairment (VI). We investigated age and cause specific prevalences of blindness and VI in a large population-based cohort in the Netherlands.
Methods: We examined 15,189 individuals aged 40 years and older using a standardized ophthalmic protocol including refraction, intraocular pressure measurement, perimetry, and imaging. All VI and blindness cases were independently reviewed by at least two investigators using all available data. Ambiguous cases were resolved in consensus meetings with senior ophthalmologists.
Results: The prevalence of VI and blindness was 2.5 percent and 0.3 percent in participants aged 40 years and older. Age related macular degeneration (AMD) was the leading cause of both blindness and VI, followed by myopic macular degeneration, glaucoma, and cataract. Cataract was an uncommon cause of blindness but the second most common cause of VI. Many cases reflected multiple coexisting conditions. The prevalence of VI decreased in more recently examined cohorts, from 9.0 percent in Rotterdam Study cohort I (participants aged 75 years and older examined in 1989) to 3.0 percent in cohort IV (2017).
Conclusion: The prevalence of blindness and VI has decreased over the last 30 years, accompanied by marked shifts in underlying causes. Persistent VI and blindness from AMD, myopic macular degeneration, and other retinal disorders highlight the need to realign resources and policy with current unmet needs.

Retinal microvasculature alterations across the Alzheimer's disease spectrum

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Introduction: Alzheimer's disease (AD) is a neurodegenerative disorder characterized by amyloid beta (Aβ) plaques, tau tangles, and vascular changes. Retinal changes associated with AD have been primarily studied in specific stages of the disease. This study aims to examine retinal microvasculature across the entire AD spectrum, including preclinical AD (pAD), mild cognitive impairment (MCI), and clinical AD, to assess the potential of retinal microvasculature as a non-invasive diagnostic biomarker.
Materials and Methods: OCTA imaging with the Zeiss plex elite 9000 was used to measure vessel density in the inner and outer ETDRS macular rings as well as capillary perfusion density and the flux index around the optic nerve head (ONH). Participants were categorized into four clinical groups based on Aβ and cerebrospinal fluid biomarkers (Healthy controls (HC) $n=84$, pAD $n=33$, MCI $n=19$, AD $n=13$).
Results: Significant reductions in the inner macular ring vessel density were found in the MCI group compared to HC ($\beta=1.65$, 95% CI: [−3.24, −0.06], $p=0.05$). The preclinical group showed a significant increase in capillary flux index compared to HC ($\beta=0.01$, 95% CI [0.001, 0.027], $p<0.05$). Perfusion density trended similarly as vessel density of the inner macular ring across the disease spectrum.
Conclusion: Retinal microvasculature alterations vary across the AD disease spectrum with the most pronounced differences observed in MCI stage. These findings highlight the potential of retinal microvasculature as a diagnostic biomarker, though larger cohort studies with longitudinal data are needed for validation.

Nurse-assisted online eye screening among older adults receiving home healthcare: Results from a cluster randomized controlled trial

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Purpose: The global population aged 65 and older is projected to double in the coming decades, increasing the prevalence of age-related vision loss. Because impaired vision can have significant consequences, we investigated the effect of an online, nurse-assisted eye-screening tool on improvements in visual acuity among older adults receiving home healthcare.
Methods: In a cluster randomized controlled trial, adults (65+) receiving home healthcare were invited. Fifty-nine home healthcare teams were randomly assigned to either the intervention or control group. Only participants in the intervention group received eye-screening, including a referral when necessary. Measurements took place at baseline, and at 6 and 12 months follow-up. Main outcomes were visual acuity (clinically relevant improvement $\geq$10 letters) and quality adjusted life years.
Results: Of 234 participants, 115 were randomized in the control group and 119 in the intervention group. Based on intention-to-treat, results showed an increased likelihood of a visual acuity improvement for the intervention group compared to controls at 12 months, but this was not statistically significant (OR: 1.58; 95%CI: 0.72–3.48; $p>0.05$). Overall, patients had a low quality of life, and QALYs did not significantly improve over time.
Conclusion: Findings suggest a non-significant clinically relevant improvement of visual acuity among older adults by using online eye-screening. Future implementation remains to be discussed in light of the current workload pressure in home healthcare and actual (eye-)health benefits in older adults.

EyeNED Platform: An open-source web-based platform for multimodal retinal imaging and AI-assisted annotation

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Purpose: The adoption of artificial intelligence (AI) for retinal image analysis has generated a demand for integrated software solutions that intuitively support their utilization and advancement. In response, we developed a vendor-independent platform designed for AI-assisted image analysis in ophthalmology.

Methods: The platform contains a fully automated image analysis pipeline, covering essential steps such as image selection, quality assessment, image preprocessing, automated segmentation of anatomic structures and lesions, and the synthesis of results in a report. The platform integrates automated image registration, enabling detailed examination of corresponding locations on multi-modal retinal images for cross-modality analysis and precise follow-up assessment. The user interface allows for navigation through images by patient ID, date, modality, annotations, or custom tags. Annotations can take the form of segmentations or forms, allowing the user to define any feature or form using a JSON schema. Structured grading is enabled through the creation of task lists.

Results: The application has been extensively used by researchers and graders within the EyeNED reading center at Erasmus Medical Center. Imaging data from 14 internal and external projects have been imported, including images from 28 distinct cameras from 6 manufacturers. Multiple AI models have been developed, including segmentation of vasculature and drusen on color fundus imaging (CFI) and retinal layers on optical coherence tomography (OCT).

Conclusions: Our software solution has demonstrated high utility in streamlining the grading process of retinal imaging. We intend to make it available as a cloud-based platform in the future, so that researchers from different centers can exchange data and algorithms.

Access to eyecare for homeless and undocumented individuals in the Netherlands

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Purpose: A growing number of uninsured homeless and undocumented individuals in the Netherlands face major barriers to accessing essential healthcare, including ophthalmic services. Despite legal entitlement to all care within the basic healthcare package, knowledge gaps among healthcare providers and fragmented local infrastructures result in care delays and avoidable morbidity. This paper aims to clarify the legal framework, quantify the affected population, identify common access problems, and provide practical tools for clinicians.

Methods: We performed a descriptive analysis of existing national data (WODC, CAK, CBS) and synthesized field experience from our work for street-medicine organizations for homeless and undocumented patients.

Results: In the Netherlands over 58,000 undocumented individuals, over 40,000 uninsured EU labour migrants, and over 30,000 homeless people lack continuous access to care. Many patients experience care refusal (up to 29%–69% in some surveys), inappropriate deferral of treatment, or administrative obstacles due to unfamiliarity with the CAK reimbursement system. Practical challenges include refusal of referrals, lack of CAK-contracted pharmacies, and provider uncertainty about the scope of permissible care. Clear criteria exist: undocumented patients are entitled to all medically necessary care within the basic package, and providers can reclaim costs through the CAK when patients cannot pay.

Conclusions: Systemic gaps in provider knowledge, administrative processes, and social infrastructure lead to significant barriers for uninsured and undocumented individuals. Improved awareness of legal entitlements, CAK reimbursement procedures, and local support networks can substantially reduce avoidable harm. These insights are essential for ophthalmologists encountering uninsured or undocumented patients in daily practice.

"I spy with my little eye": Charles Bonnet syndrome in Stargardt disease and the broader low-vision population

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Purpose: Charles Bonnet syndrome (CBS) is characterized by complex visual hallucinations in individuals with

visual impairment in the absence of psychiatric disease. Although traditionally considered an age-related condition, increasing evidence suggests that CBS occurs across all age groups with acquired vision loss. We set out a study to determine the prevalence and risk factors of CBS in patients with Stargardt disease (STGD1; Dhooge PPA et al. Br. J. Ophthalmol. 2023;107 (2):248–253). These findings are interpreted within the context of the broader low-vision population.
Methods: Eighty-three Dutch patients with STGD1 were screened for CBS. Patients with suspected CBS participated in structured interviews. Results were interpreted alongside data from recent systematic reviews and meta-analyses on CBS in low vision patients.
Results: CBS prevalence in our STGD1 cohort was 8.4%, with a female predominance. CBS was associated with reduced social functioning and moderate visual impairment but not with age or retinal atrophy extent. Hallucinations typically developed following a decline in visual acuity. These findings are consistent with meta-analytic evidence showing a pooled CBS prevalence of approximately 20% among low vision patients, indicating that CBS risk relates more strongly to severity and recent onset of vision loss than to age or underlying diagnosis.
Conclusions: CBS is a frequent yet underrecognized consequence of visual impairment, including in younger patients with STGD1. Routine screening for CBS in all patients with moderate to severe vision loss may improve recognition, reduce anxiety, and enhance patient care.

Patient reported outcome measure and efficiency of an asynchronous virtual ophthalmic clinic

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Purpose: Patient reported outcome measures are crucial to include in the evaluation of cure and care, in addition to clinical and health economy data. Our aim was to evaluate asynchronous (data collection and evaluation at different points in time) virtual clinics, by patient reported outcome measures and efficiency.
Patients and methods: In 949 patients who visited an asynchronous virtual clinic, we recorded patient satisfaction with a 4-point scale ranging from very unhappy to very happy, and by a free-text response field. In addition, we measured patient throughput time, from check-in to departure. The time spent at each diagnostic test was measured separately.
Results: 97% of patients were happy or very happy. The most mentioned patient remark was “rapid/quick”. On average patient throughput time was 28 min.
Conclusions: Patients appear to be satisfied with asynchronous ophthalmic clinics, especially with the reduced time they spent per visit.

Evaluation of low luminance visual acuity in retinitis pigmentosa: relationship with full-field stimulus testing and visual acuity: REPEAT study report no. 5

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Purpose: Reliable functional outcome measures are essential for detecting subtle changes in visual function in retinitis pigmentosa (RP), especially in early stages when BCVA remains preserved. This study assessed the applicability and reliability of low luminance visual acuity (LLVA) as a clinical outcome measure in a genetically heterogeneous RP cohort by evaluating its correlation with full-field stimulus test (FST) and BCVA.
Methods: Data were obtained from the prospective observational REPEAT cohort. We included 49 patients (49 eyes) with clinically diagnosed RP, including 45 genetically confirmed cases. BCVA and LLVA were measured using ETDRS charts, with LLVA assessed through a 2-log neutral density filter. Low luminance deficit (LLD) was calculated as the difference between LLVA and BCVA. FST was performed in a subgroup with BCVA of 0.40 logMAR or worse, using chromatic rod- and cone-targeting stimuli. Spearman correlation assessed associations between LLD, FST, BCVA and LLVA.
Results: LLD showed strong correlations with both rod- and cone-associated FST thresholds (blue: rho = 0.626, $p<0.001$; red: rho = 0.581, $p<0.001$). Median BCVA was relatively preserved (56 ETDRS letters), while median LLVA was substantially reduced (35 ETDRS letters). LLVA correlated strongly with BCVA (rho = 0.910, $p<0.001$).
Conclusions: LLVA identifies clinically relevant functional deficits not detected by BCVA, particularly in early RP. The strong relationship with FST shows LLVA captures central and peripheral visual dysfunction. LLVA is a reliable and sensitive complementary outcome measure to FST in clinical practice and research.

Positioning Mendelian myopia at first presentation on population axial length centiles

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Purpose: Mendelian myopia (myopia caused by single-gene mutations) is characterised by early onset and rapidly progressive axial elongation, leading to extreme refractive error. This study compared axial length (AL) at initial presentation across Mendelian disorders with population-based growth curves to evaluate prognostic utility.

Patients and Methods: Genetically confirmed Mendelian myopia patients were identified from electronic records at two Dutch tertiary centres. AL, spherical equivalent (SER), age, and sex were extracted. Patients were grouped into five subgroups: retinal signaling disorders (*CACNA1F, NYX, GRM6, GPR179, TRPM1, OPN1LW, PDE6H*), primary photoreceptor dystrophies (*RPGR, GUCY2D*), Stickler syndrome (*COL2A1, COL11A1, COL9A1*), Knobloch syndrome (*COL18A1*), and other monogenic forms (*ARR3, PAX6, SLC39A5*). Per subgroup, AL at first presentation was compared with age- and sex-specific 98th-percentile values from population-based AL curves to determine the proportion exceeding this threshold and mean excess AL.

Results: 59 patients were included; 34 (24 males) had ≥1 AL measurement. Median age at first AL measurement was 6.8 years (range 6.0–18.5). AL exceeded the 98th percentile in 9/10 patients with retinal signaling disorders (mean excess 1.45 mm ±1.13), 2/2 photoreceptor dystrophies (0.94 mm ±0.68), 15/16 Stickler syndrome (2.53 mm ±2.48), 2/2 Knobloch syndrome (3.10 mm ±1.44) and 2/3 other monogenic forms (4.22 mm ±4.49).

Conclusion: AL in Mendelian myopia exceeds the 98th population centile at first presentation, with substantial variation across subtypes. Consequently, population-based centile charts are not suitable for characterizing these extreme phenotypes, highlighting the need for gene-specific growth curves. Systematic AL measurement remains essential for accurate prognosis and follow-up.

Recognizing retinitis pigmentosa genotypes and phenotypes using adaptive optics imaging

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Purpose: To characterize retinitis pigmentosa (RP) using adaptive optics flood illumination ophthalmoscopy (AO-FIO), focusing on cellular and topographic retinal phenotypes.

Methods: AO-FIO was conducted using a protocol covering 4° nasal to 12° temporal and −5° to +5° vertically. Montages were evaluated for cellular phenotypes (cone visibility, mosaic regularity, hypo-reflective cones, and retinal pigment epithelium (RPE) cell visibility) and topographic phenotypes (hyper/hypo-reflective pigmentation patterns, atrophic or blurred parafoveal rings, regional reflectivity abnormalities, and choroidal visibility) across foveal (1°), perifoveal (1–4°), and parafoveal (4–10°) eccentricities. AO-FIO findings were co-localized with OCT for interpretation.

Results: In twenty RP patients (37 eyes; median age 37 years [IQR: 29–48]; median BCVA 0.75 [IQR: 0.50–0.90]), cones were visible in more than 80% of the eyes, all showing irregular spacing. Hypo-reflective cones ($n=11$) appeared in the perifovea and corresponded to ellipsoid/interdigitation zones (EZ/IZ) disruption and outer nuclear layer (ONL) thinning on OCT. RPE mosaics ($n=3$) occurred only in regions of severe outer retinal loss. Pigmentation abnormalities ($n=16$) included homogeneous reflectivity changes and localized atrophic clumps. Parafoveal rings ($n=7$) were blurred ($n=4$) or atrophic ($n=3$), independent of image quality or cystoid macular edema. Choroidal vessels ($n=9$) indicated advanced ONL, EZ/IZ, and RPE loss. Homogeneous pigmentation changes and choroidal visualization occurred most frequently in *USH2A*-associated RP.

Conclusions: AO-FIO enables multi-scale phenotyping of RP by revealing both cellular abnormalities and topographic patterns. These phenotypes correlate with structural OCT changes and demonstrate the potential of AO-FIO for detecting degeneration, and supporting biomarker development for future clinical trials.

Seven-year prospective follow-up in patients with *CRB1*-associated retinal dystrophies

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Purpose: To describe the natural disease course of CRB1-associated retinal dystrophies over a seven-year period to aid in the design of endpoints for future clinical trials. Previous analyses showed a decline in macular sensitivity, best-corrected visual acuity (BCVA) and visual field. This extended seven-year analysis aims to quantify the long-term rate of progression and validate the most sensitive progression markers.

Methods: This single-center, prospective case series primarily included 22 patients with CRB1-associated retinal dystrophies. Patients underwent ophthalmic assessment at baseline, 2, 4, and 7 years after baseline. Clinical examination included BCVA using ETDRS charts, Goldmann kinetic perimetry (V4e isopter), microperimetry, full-field stimulus test (FST), and spectral-domain OCT. The 7-year data are currently being collected and analysed using a longitudinal model to determine the long-term progression of functional and structural outcomes.

Results: The analysis of the 7-year follow-up data is currently underway to quantify the long-term rate of progression. The final quantitative 7-year results and accompanying measures of significance will be available for presentation at the conference.

Conclusions: Except for FST responses, a significant decrease was observed over four years in the visual function parameters of BCVA, Goldmann perimetry, and microperimetry. These measures may thus be used as sensitive clinical endpoints. The 7-year results are expected to confirm the long-term progression rates of these parameters, crucial for the design of future human treatment trials.

Feasibility and promise of paired-eye trial designs in *USH2A*-related retinitis pigmentosa

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Purpose: In the light of gene therapy trial design, paired-eye models, where one eye is treated and the other serves as a control, become increasingly relevant. We analyse best corrected visual acuity (BCVA) and retinal sensitivity (RS) measured by static perimetry (SP) in *USH2A*-related retinitis pigmentosa (*USH2A*-RP) patients, and estimate the sample sizes required to support paired-eye designs in future clinical trials.

Methods: A total of 36 participants participated in the Characterizing Rate of Progression in USHer Syndrome (CRUSH) study. They underwent annual SP and BCVA measurements over four years. Annual rates of change were calculated, inter-eye symmetry was examined using intraclass correlation coefficients (ICCs) and linear mixed models, and sample sizes for paired-eye versus traditional trial designs using SP and BCVA were estimated.

Results: SP declined by −0.56 decibels/year (95% CI: −0.68, −0.44; −11.6%). BCVA worsened by 0.008 LogMar/year (95% CI: 0.002, 0.014; −1.7%). Inter-eye symmetry was excellent for SP (ICC: 0.80 (95% CI: 0.63, 0.90)) but poor for BCVA (LogMar) (ICC: 0.37 (95% CI: 0.05, 0.62)). For a future four-year trial, sample size estimates showed an advantage of paired-eye designs: 167 versus 18 for SP, and 2160 versus 689 for BCVA.

Conclusions: This study found that SP consistently declined with strong inter-eye symmetry, while BCVA showed variable progression. SP emerged as a promising endpoint for paired-eye trials, combining faster detection of progression and substantially lower sample sizes. These findings support the use of paired-eye designs with SP as an endpoint in future *USH2A*-RP clinical trials.

Toward a multimodal imaging classification system for inherited retinal dystrophies (IRD)

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Purpose: Growing momentum in therapeutic research and advances in AI-driven quantitative retinal imaging

analysis underscore the necessity of well-defined structural outcome measures for rod–cone inherited retinal dystrophies (IRDs). We propose an expert consensus–driven multimodal IRD classification system to meet this growing need.
Methods: Lesion components characteristic of rod-cone IRD and their imaging correlates were identified through a review of the existing literature and exploratively assessed in multimodal imaging data (OCT, CFI, FAF) from patients registered in the RD5000 database (USH2A: 27 patients/54 eyes; EYS: 9 patients/18 eyes) supplemented with an extra set of images from The Rotterdam Eye Hospital (RPE65: 21 patients/33 eyes). Representative images were first reviewed by two ophthalmologists and one imaging expert, followed by pixel-level annotation by two graders using the in-house developed EyeNED platform. Features with substantial interpretative disagreement were flagged for targeted discussion.
Results: We selected patients with multimodal imaging data from at least two consecutive follow-ups (USH2A: 20 patients/40 eyes; EYS:8 patients/16 eyes; RPE65: 21 patients/33 eyes) for this study. In early IRD, OCT-derived structural changes and FAF features (e.g., hyperautofluorescent ring) were readily delineated with high intergrader agreement. In advanced IRD, particularly recognition of features as well as lesion boundaries showed less intergrader agreement. In addition, the distinction between stages of hypoautofluorescence on FAF was considered challenging.
Conclusions: We present initial steps toward a standardized multimodal imaging classification system for rod-cone IRD by identifying and addressing interpretative discrepancies in imaging correlates. These efforts will be consolidated in an upcoming expert consensus meeting, with the aim of establishing an internationally accepted classification scheme.

Pseudo-Stargardt retinal dystrophy associated with *PRPH2* gene variants

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Introduction: *PRPH2*-associated pseudo-Stargardt retinal dystrophy (PPRD) is an inherited retinal dystrophy associated with variants in the *PRPH2* gene, which can cause a wide range of clinical phenotypes and severity. The pathophysiological mechanisms that underlie PPRD, and the reasons for this wide phenotypic variation, remain poorly understood.
Methods: In this cohort study we included 109 PPRD patients and collected data on visual acuity, visual field exams, and electrophysiology. Imaging data included color or pseudocolor fundus images, fundus autofluorescence, infrared imaging, and spectral-domain optical coherence tomography (SD-OCT). We analyzed the clinical spectrum, natural history, and genotype-phenotype correlations of PPRD.
Results: We identified four disease stages of PPRD based on multimodal imaging. In stage 1, the retina appeared normal on fundoscopy and OCT, with mild foveal abnormalities visible only on near-infrared reflectance imaging. In stage 2a, isolated multifocal yellow-white flecks were observed in the macula or near the temporal vascular arcades. In stage 2b, these flecks were visible both in the macula and near the vascular arcades. In stage 3, these flecks became confluent, expanding outward over time and eventually covering the macula, with areas of depigmentation, mild atrophy and decreased autofluorescence. In stage 4, well-demarcated chorioretinal atrophy emerged, either outside (stage 4a) or within the fovea (stage 4b).
Conclusions: PPRD is an autosomal dominant progressive disease that can mimic other hereditary retinal diseases such as Stargardt disease. Despite its wide variability in severity, up to 25% of patients with PPRD can develop severe vision loss starting in the fifth decade of life.

Managing macular sub-internal limiting membrane hemorrhage without surgical intervention

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Purpose: Sub-internal limiting membrane (sub-ILM) hemorrhages have diverse causes and may lead to sudden vision loss. Evidence on prognosis is limited, and management strategies vary. This study investigates the natural course, visual outcomes, anatomical recovery, and the potential need for surgical intervention in sub-ILM hemorrhages. To date, there is little evidence on their prognosis, and proposed management approaches are variable.
Methods: Thirteen patients with sudden vision loss due to fovea-involving macular sub-ILM hemorrhage were followed longitudinally with optical coherence tomography (OCT) and visual acuity (VA) measurements until complete resolution.
Results: Presumed causes included Valsalva retinopathy (n=4), thrombocytopenia (n=3), and unknown origin (n=6). In patients with measurable hemorrhage on OCT and follow-up >4 weeks, spontaneous regression occurred within 12 weeks, with final LogMAR visual acuity <0.10 (Snellen > 0.8). Two patients had unmeasurably large hemorrhages (>1500 µm) of one required vitrectomy with ILM peeling due to dense vitreous hemorrhage; the other resolved conservatively over 16 weeks. At the final OCT, mean central subfield thickness (CST) was significantly higher in eyes after resolution of the sub-ILM hemorrhage compared with unaffected fellow eyes (CST: 300 µm vs 279 µm, p < 0.001).
Conclusion: Sub-ILM hemorrhages measurable on OCT typically resolve spontaneously with good visual outcomes. Vitrectomy should be considered in severe cases, such as dense vitreous hemorrhage or extremely large hemorrhages, though conservative management may still succeed. These findings provide practical guidance for clinicians in the management of sub-ILM hemorrhages. Longer-term studies are needed to assess the risk of epiretinal membrane formation and factors influencing hemorrhage clearance.

Intra-ocular inflammation following intravitreal faricimab injections: A retrospective case series

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Purpose: To report the incidence and clinical course of intraocular inflammation (IOI) following intravitreal faricimab injections (IVF).
Methods: This retrospective case series consists of 262 eyes of 223 patients. The patients were treated at our tertiary care center between March 2023 and April 2025 with IVF for the following indications: neovascular age related macular degeneration (n = 144), diabetic macular edema (n = 59), macular edema secondary to retinal vein occlusion (n = 6), and other types of choroidal neovascularisation (n = 14).
Results: A total of 1732 IVF injections were administered. Ten eyes of seven patients developed IOI. The incidence of acute IOI at our clinic was 3.8% per eye and 0.6% per injection. Main complaints at presentation were floaters and blurred vision. In all cases the inflammation was relatively mild and responded well to topical and/or subconjunctival steroids. There were no cases of vasculitis. Four patients were treated with topical steroids, one patient with subconjunctival steroids, and two patients with both topical and subconjunctival steroids. The interval between last IVF and onset of IOI varied from 2 to 42 days (mean 18 days).
Conclusion: Our study shows an incidence of IOI of 3.8% per eye and 0.6% per injection. These results are in line with those reported in phase III clinical trials. Our cases of IOI were mild and responded well to local steroids and the visual acuity fully recovered. It is recommended to question the patients and be vigilant in cases of new floaters or sudden blurred vision following IVF, which could indicate an IOI.

Automated assessment of diabetic retinopathy using AI: Feasibility in daily practice

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Introduction: Since 2020, the Jeroen Bosch Hospital has acted as the coordinating center for integrated diagnostics in the Northeast Brabant region. This has led to a substantial rise in the number of fundus photographs submitted for diabetic-retinopathy screening. To manage the increasing workload, we evaluated whether automated AI-based assessment could reduce the number of images requiring manual review. RedCAD (Thirona) was selected following a prior evaluation process.
Patients and Methods: A six-month pilot study was performed to assess the quality and clinical reliability of automated grading. Between January 5 and June 29, 2025, a total of 1376 examinations were reviewed independently by an optometrist and the RedCAD algorithm, without the optometrist having access to AI results. In 102 examinations, one or both eyes were ungradable for the optometrist, while 101 examinations lacked an AI outcome. Consequently, 1196 examinations were included in the final analysis.
All cases in which RedCAD and the optometrist reached different conclusions were subsequently assessed by an ophthalmologist, who provided the reference standard.
Results: Across all evaluated eyes, 91% showed identical grading between the algorithm and the optometrist. Agreement on referral decisions was 97.1%. The

algorithm achieved a sensitivity of 92% and specificity of 99.1% for detecting referable diabetic retinopathy.

Conclusion: Automated assessment with RedCAD did not appear to miss additional required referrals compared with optometrist review. Although the algorithm tended to score mild abnormalities (R1) somewhat higher, the number of false-positive referrals remained limited. These findings suggest that AI-assisted evaluation may be a feasible tool to support routine diabetic-retinopathy screening and reduce clinical workload.

Real-world 24-month treatment outcomes of aflibercept versus bevacizumab for neovascular age-related macular degeneration

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Purpose: Dutch guidelines recommend bevacizumab as first-line treatment for neovascular age-related macular degeneration (nAMD) due to its lower cost and comparable outcomes to other anti-VEGF agents. However, Dutch clinics report higher injection frequencies than international counterparts, raising concerns about treatment burden. This pilot study compares 24-month real-world outcomes of initiating aflibercept versus bevacizumab in Dutch community ophthalmology clinics.

Methods: An observational study using prospective data from three Dutch FRB registry clinics was conducted. Two clinics used bevacizumab with two-weekly T&E interval adjustments. A third clinic initiated aflibercept with a four-week initial extension, followed by two-weekly adjustments. Primary outcomes were total injections and injections to CNV inactivity. Secondary outcomes included time to CNV inactivity, VA change, and final injection interval.

Results: We included 43 eyes in the aflibercept group and 129 eyes in the bevacizumab group, matched 1:3 on age and baseline VA. At 24 months, mean VA was 65.6 (aflibercept) vs 68.8 (bevacizumab) letters ($p=0.182$), and mean letter gains were +0.8 vs +4.2 ($p=0.113$). Aflibercept-treated eyes received 5.5 fewer injections ($p<0.001$), required 5.6 fewer injections to achieve CNV inactivity ($p<0.001$), and reached inactivity 20.6 weeks earlier ($p=0.001$). Final injection intervals were 3.3 weeks longer with aflibercept ($p<0.001$); 41.9% reached intervals ≥12 weeks compared to 14.7% in the bevacizumab group ($p<0.001$).

Conclusions: Aflibercept under a modified T&E regimen resulted in fewer injections, faster disease control and longer treatment intervals, while VA was not significantly different.

Long-term follow-up: Direct versus crossover to half-dose photodynamic therapy for chronic central serous chorioretinopathy

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Introduction: The PLACE and SPECTRA trials demonstrated superiority of half-dose photodynamic therapy (PDT) over high-density subthreshold micropulse laser (HSML) and eplerenone, respectively, for treatment of chronic central serous chorioretinopathy (cCSC). This study evaluates long-term outcomes in patients randomized to PDT directly versus after crossover.

Patients and Methods: This multicenter retrospective study included 179 cCSC patients from the PLACE and SPECTRA trials who received half-dose PDT within 12 months of enrollment at a Dutch medical center. Of these, 107 patients (60%) returned for long-term follow-up at a median of 7.2 years (IQR: 6.2–8.8) after their first PDT. Patients were categorized as direct (initial randomization to PDT) or delayed PDT (crossover after HSML or eplerenone).

Results: At long-term follow-up, 91.6% of patients showed complete resolution of subretinal fluid (SRF): 88.3% in the direct PDT group vs. 95.7% in the delayed PDT group ($p=0.293$). SRF recurrence occurred in 20.0% of direct PDT vs. 38.3% of delayed PDT patients ($p=0.097$). Best-corrected visual acuity (BCVA) improved significantly in both groups: +7.6 ETDRS letters after direct PDT ($p<0.001$) and +3.4 letters after delayed PDT ($p=0.031$); the adjusted between-group difference favored direct PDT (+4.2 letters, $p=0.002$).

Conclusion: PDT provides durable anatomic and functional benefits in cCSC for more than 7 years. Early PDT results in greater BCVA gains and may lower recurrence risk compared with delayed PDT. Delayed PDT remains

effective, supporting the long-term safety and robustness of PDT as the preferred therapy for cCSC.

Screening of genes associated with inherited retinal dystrophies in patients with a primary diagnosis of age-related macular degeneration

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Purpose: Assessing the prevalence of IRD variants in AMD patients and re-evaluating the phenotype of individuals carrying these variants.
Methods: A total of 1956 unrelated probands (1115 previously diagnosed as AMD, 841 controls) were recruited. Next generation sequencing was used to screen 105 genes associated with IRDs and AMD. Additionally, 52 AMD-associated variants were used to calculate the AMD genetic risk score (GRS), which was then classified as low, normal, or high. For individuals carrying a pathogenic variant in an IRD gene, multimodal imaging was re-evaluated to determine if the phenotype was consistent with an IRD, remained classified as AMD or control, or was inconclusive.
Results: Of the 1956 probands, 35 (1.8%) carried a putative IRD variant. After excluding 12 previously described cases, 23 individuals remained (20 with primary AMD and 3 controls). Among the 20 AMD patients, six retained the AMD diagnosis, six were reclassified as IRDs, and two had a hybrid phenotype (Best disease + AMD, Stargardt + AMD). Nine others were inconclusive or remained controls. Rare complement variants were found in two individuals (one AMD patient and one control). A high GRS was observed in 88.9% of AMD patients (with or without concurrent IRD) compared to 33.3% of those diagnosed with an IRD without concurrent AMD.
Conclusion: Identification of IRD variants in AMD patients may lead to reclassification of their phenotype, highlighting the difficulty in differentiating between the two conditions. Our findings suggest that genetic screening can help accurately diagnose ambiguous cases.

Oral plus intravitreal treatment for viral retinitis: The Amsterdam experience

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Purpose: Here we report all 28 cases of viral retinitis since the introduction in 2018 of oral antiviral treatment as an alternative for intravenous treatment in a large teaching hospital in the Netherlands.
Methods: Of 8 out of 10 cytomegalovirus-positive (CMV) patients with 15 affected eyes treated primarily with oral antiviral medication, only 1 was switched to intravenous medication. Of 18 patients with a diagnosis of varicella zoster virus (VZV) ARN, 16 were initially treated orally, while 2 were started on intravenous antivirals. All but 1 CMV and 1 VZV patient were treated with a median number of respectively 1.5 and 4.0 intravitreous foscarnet injections: 1 CMV patient did not get any injections, while 1 VZV patient received 4 ganciclovir injections.
Results: Outcome after a mean of 17 months of follow up was mostly favorable in the CMV group, with only 2 out of 15 eyes reaching a final vision of 0.4 LogMAR or worse while in the VZV group, 8 out of 20 affected eyes had a vision of 0.4 LogMAR or worse, with 3 out of 20 having a final vision worse than 0.8 LogMAR and 2 eyes in two patients having no light perception. A total of 4 out of 35 eyes developed a retinal detachment.
Conclusion: Overall, the outcomes of this retrospective cohort of subsequent viral retinitis patients treated after the introduction of oral antiviral medication with complimentary intravitreous injections seem at least comparable with the reported outcomes in the age of primary intravenous treatment.

5-year retinal vein occlusion outcomes – An FRB! analysis

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Purpose: To evaluate 60-month outcomes of eyes treated with vascular endothelial growth factor (VEGF) inhibitors for retinal vein occlusion (RVO). In this study, our clinic starting with bevacizumab is compared with other clinics using the Fight Retinal Blindness! registry.
Methods: Multicentre, international, observational study with data collected from 12 countries, tracking 496 treatment-naïve BRVO eyes and 474 treatment-naïve CRVO/HRVO eyes between 2016 and 2024. The focus is on the completers of every 12-month period.
Results: In BRVO, the mean visual acuity in our clinic was significantly higher than in other countries (e.g. 60-month completers 81.7 versus 77.0 letters, respectively) but no significant differences were found in VA change over the 60-month period (e.g. 20.6 versus 15.0 letters, respectively). Our clinic needed significantly more injections during every year (for example 5-year completers 39.1 versus 24.8 injections, respectively) and inactivity was significantly different from year 2 onward leading to 94% inactivity in the fifth year versus 54% in other countries. In CRVO/HRVO, the mean visual acuity in our clinic was significantly higher than in other countries (e.g. 60-month completers 75.7 versus 58.6 letters, respectively) but no significant differences were found in VA change over the 60-month period (e.g. 20.8 versus 10.5 letters, respectively). Significantly more injections were given every year in our clinic (e.g. 5-year completers 41.2 versus 24.2 injections, respectively). At 5 years, our inactivity percentage in CRVO/HRVO was not significantly higher: 78% versus 56%.
Conclusion: After 5 years, BRVO patients in our clinic received significantly more injections, which resulted in preservation of their significantly higher visual acuity and more inactivity. Our CRVO/HRVO patients

performed excellent with a significantly higher visual acuity at the cost of significantly more injections. Our intensive treatment in RVO results in outstanding visual outcomes after 5 years.

Neovascular age-related macular degeneration without drusen

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Purpose: To describe the clinical characteristics and one year follow-up of patients with neovascular AMD (nAMD) without drusen in either eye.

Methods: This is a multicentre retrospective cohort study in 3 tertiary referral centres in the Netherlands. We included patients of 55 years or older with nAMD in one or both eyes, without the presence of drusen or signs of another underlying disease in either eye. The medical charts and multimodal imaging (MMI) were evaluated by 2 independent graders. Eyes were divided into 2 groups: polypoidal choroidal vasculopathy (PCV)-associated macular neovascularization (MNV), and non-PCV MNV. We evaluated the visual acuity (VA) at baseline and 1 year after baseline; complete resolution of macular fluid on optical coherence tomography (OCT) during 1 year of follow-up; required treatments to achieve a complete resolution of macular fluid on OCT; complication rate.

Results: We included 106 eyes of 99 patients, with a median age of 73 years. Seventy-one eyes had PCV-associated MNV, and 35 eyes had non-PCV MNV. Overall median baseline VA was 0.22 logMAR (Snellen 20/30), and 0.15 logMAR (Snellen 20/28) at 1 year follow-up. In total, 31 out of 48 eyes (65%) achieved complete resolution of macular fluid on OCT during follow-up, most commonly achieved with combined photodynamic therapy (PDT) + intravitreal anti-vascular endothelial growth factor (anti-VEGF) injections (52% of all eyes achieving complete resolution).

Conclusions: nAMD can occur without the presence of drusen in either eye. Treatment with anti-VEGF injections, PDT, or a combination thereof appears effective, but more research is warranted.

Prevalence of inflammatory features in myopic choroidal neovascularization

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Purpose: To investigate the incidence of inflammatory imaging biomarkers in patients with myopic choroidal neovascularization (mCNV).

Methods: This retrospective review of medical records and imaging analyzed data of patients diagnosed with mCNV who received anti-VEGF injections between 2017 and 2024 in the Rotterdam Eye Hospital. All imaging modalities (color fundus photographs, OCT and OCT-angiography scans) were graded by the first author. Scans with signs of inflammation were reviewed by two uveitis and medical retina specialists.

Results: In 100 eyes from 51 patients evaluated thus far, CNV was identified in 63 eyes, of which 60 eyes received an average of 10 (median 4.5, IQR: 12) anti-VEGF injections per eye. In the analysis, 29 eyes from 17 individuals demonstrated signs of inflammation. 22 eyes were diagnosed with CNV, of which 21 eyes underwent anti-VEGF treatment with a mean of 12 injections per eye (median 6, IQR: 15.5). 7 eyes were contralateral eyes without CNV. Patients in the inflammation group were younger at the time of diagnosis (51 vs. 61 years (mean), $p<0.04$). Eyes exhibiting inflammation also showed more severe myopia, although it did not reach statistical significance (SE: −11.4 D vs −13.5 D, $p=0.08$).

Conclusions: In 100 eyes of patients, of which 63 eyes were diagnosed with mCNV, 29% showed signs of inflammation. This evidence provides new insights into pathophysiology of mCNV, motivating further research into therapeutic opportunities addressing the inflammatory component, possibly decreasing the incidence of CNV's or the number of injections needed.

Adjuvant Navilas™-laser in BRVO: A happy end to endless injections?

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Introduction: The Branch retinal vein occlusion Study (BVOS) showed benefit of grid laser in eyes with visual acuity (VA) <5/10 Snellen. However, intravitreal anti-VEGF showed superior VA outcome, and laser was discarded by many. With Navigated laser (Navilas™) allowing more precise image-guided coagulation, a combination of laser/injections could be a modality to optimize VA and reduce the injection burden.

Methods: In 167 eyes treated with a combination of anti-VEGF injections and either early (within 6 months, $n=69$), or deferred Navilas™ (after 6 months, $n=98$) VA (Snellen), OCT (CFT, um) and number of injections were studied for 3 years post-laser.

Results: VA at onset of BRVO was 0.4±0.2. Post-laser VA was 0.7±0.3 at 1 year, 0.7±0.3 at 2 and 0.7±0.3 at 3 years. CFT (at onset 450±280 µm) decreased to 310±180 µm post-laser. In eyes with deferred laser, the injection burden was 7±5 injections in the year prior to laser. The number of injections dropped to 5±3 injections in year 1, 3±3 in year 2 and 3±3 in the 3rd year post-laser ($p<0.05$). In 23% of eyes injections were stopped along year 1 post-laser, an additional 16% along year 2. Between early or deferred cohorts, there was no significant difference in post-laser number of injections or rate of injection-free eyes.

Conclusions: In BRVO, adjuvant Navilas™ laser seems to reduce the number of anti-VEGF injections significantly, and in 39% of eyes resulted in stopping injections altogether. As adjuvant laser appears not to decrease VA, it could be considered to reduce the injection burden.

Focus- and roll-off-dependent correction for attenuation coefficient estimation in optical coherence tomography

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Introduction: Traditionally, retinal OCT focuses on layer thickness and structure, but OCT signal intensity is emerging as a potential biomarker. Interpretation, however, is complicated by biases due to optical effects. Deriving attenuation coefficients (AC) from signal intensities is a major step in reducing bias. The AC describes intensity decrease as light travels through tissue, and allows detection of subtle changes, such as reduction of AC in the nerve fiber layer reported in glaucoma (Chang et al., 2022). Yet, the AC is still affected by the OCT-scanner's focus settings and depth sensitivity decay (roll-off). This study aims to improve AC reproducibility and reliability, enabling more robust longitudinal assessment of retinal changes.

Materials and Methods: A correction model was developed to account for focus depth and roll-off, extending from a previously published focus-only approach by Kübler et al. (2021). The method was evaluated through simulations at different focus and axial positions. Quantitative assessment included layer-wise mean and standard deviation of AC values and the root-mean-square error (RMSE) before *and* after correction.

Results: Combined focus and roll-off correction reduced bias in AC values. Corrected data showed improved consistency, lower RMSE, and decreased variability across layers compared with uncorrected measurements. The approach remained robust under varying noise levels, indicating stable performance across imaging conditions.

Conclusion: Incorporating focus-depth and roll-off corrections improves the precision and reproducibility of OCT-based AC estimation. Further validation in phantom and *in vivo* data should clarify its clinical utility, potentially enabling more reliable monitoring of retinal disease progression.

Evaluating the 'macular degeneration: What happens next?' Question prompt list: Enhancing communication in ophthalmology consultations

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Purpose: Effective communication is essential in healthcare, as it builds trust between patients and providers, improves understanding of diagnoses and treatment plans, and reduces patient anxiety. Age-related Macular Degeneration (AMD) is a complex condition where clear communication between patients and healthcare providers (HCPs) is particularly important. This study evaluates the communication tool 'Macular Degeneration: What Happens Next?', a Question Prompt List (QPL) designed to help patients formulate questions and support HCPs in guiding consultations.

Methods: The development process included a literature review and surveys to gather insights into patients' experiences, needs, and preferences. These findings informed focus group discussions with patients, informal caregivers, and healthcare professionals to shape the communication tool. The QPL was further refined through cognitive interviews. The tool's impact was evaluated using video-recorded consultations between patients and ophthalmologists ($n=51$). Post-consultation surveys assessed patients' communication experiences and evaluated the QPL.

Results: Patients and informal caregivers in the QPL group asked more questions (7.5 vs. 5.7 per consultation) and introduced more new discussion topics (3.8 vs. 1.4) than those in the control group, indicating more active communication during consultations. Patients using the QPL also reported higher confidence in communicating with their clinician, with consistently higher scores across all assessed communication domains. The QPL itself was evaluated positively, receiving an average rating of 7.6/10, and all participants (100%) indicated they would recommend it to others.

Conclusions: The QPL demonstrates potential to enhance patient-provider communication. It may further provide a foundation for creating similar tools for other ophthalmic conditions.

Two-year results of switching to faricimab in refractory exudative age-related macular degeneration

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Purpose: to evaluate the two-year outcomes of switching to faricimab in refractory exudative age-related macular degeneration (AMD).
Methods: Consecutive case series.
Primary outcome measurement: Injection interval.
Results: Between April 2023 and April 2024, 115 eyes from 115 patients with exudative AMD, previously treated with at least two different intravitreal anti-VEGF agents (bevacizumab, ranibizumab, aflibercept and/or brolucizumab), were switched to faricimab injections. Reasons for switching were: unsatisfactory reduction of (sub)retinal fluid and/or inability to extend injection intervals beyond 4 weeks. After switching to faricimab, the injection interval increased by 1.8±2.1 weeks (mean±SD) at 12 months and by 2.1±2.4 weeks at 24 months compared to baseline (p=0.002). An injection interval of 12 weeks or more was achieved in 2% of patients at 12 months and in 6% at 24 months. Visual acuity changed 2±6 letters at 12 months, and 1±15 letters at 24 months (p=0.066). Central retinal thickness decreased -53±108 µm at 12 months and -60±112 µm at 24 months (p<0.001). Of the 115 patients, 35 were lost to follow-up: 8 patients had died, 4 discontinued treatment due to the high burden, and 23 patients (22%) were switched back to a previous anti-VEGF agent because of insufficient effect on (sub)retinal fluid and/or a decline in visual acuity. Faricimab was discontinued in one patient due to an inflammatory reaction.
Conclusion: When switching to faricimab in refractory exudative ARMD patients on average a 2 week gain in injection interval can be expected in the majority of patients.

Lutein and zeaxanthin in AMD supplements: A crossover bioavailability trial and evaluation of biochemical contents in commercial formulations

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Purpose: AREDS2-based supplements may help reduce the risk of advanced age-related macular degeneration. We evaluated serum L&Z levels after intake of two formulations. Additionally, we analyzed the AREDS2-relevant nutrient content of 23 commercial supplements to assess label accuracy and overall formulation quality.
Methods: In a randomized, crossover, open-label study, 11 healthy adults, mean age 41 years (SD: 12.71) received either an AREDS2 soft gel capsule (B1) or an herbal-coated tablet (B2) for four weeks. Serum lutein and zeaxanthin concentrations were measured at baseline and after intervention. 23 supplements were analyzed for lutein, zeaxanthin, vitamins C and E, zinc, copper, and omega-3 fatty acids content. Measured values were compared with product label claims and with AREDS2 reference levels.
Results: B1 supplementation produced significant increases in serum lutein (0.13→0.31 µmol/L, p=0.003) and zeaxanthin (0.08→0.11 µmol/L, p=0.003). In contrast, B2 supplementation had no measurable effect, which may be attributable to the herbal-coating allowing oxidation. Analysis of commercial supplements revealed substantial variability in nutrient content: measured lutein ranged from 75.7% to 206.9% of labeled amounts, and zeaxanthin from 58.7% to 326.0%. Only 5 of the 23 products contained AREDS2-recommended levels of key nutrients. Vitamins C and E, zinc, copper, and omega-3 fatty acids content varied widely, with some nutrients undetectable in certain products.
Conclusion: Soft gel capsules containing AREDS2 formula significantly enhanced lutein and zeaxanthin serum levels. The variability in commercial supplement contents highlights the need for quality control and careful product selection when aiming to meet AREDS2 formula standards.

Late-onset Stargardt: Characteristics, genetics, and progression

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Purpose: We describe the disease characteristics, underlying genetics, and disease progression of late-onset Stargardt disease (STGD1), a subtype of STGD1 defined by onset of symptoms at 45 years or older, and highlight the differences from geographic atrophy.
Methods: In this retrospective cohort study, we reviewed the medical records of 71 patients with late-onset STGD1, including initial symptoms, visual acuity, and retinal

findings. Fundus autofluorescence images and OCT scans were quantitatively and qualitatively assessed.
Results: Median age at onset was 55 years (range, 45–82). The most common genetic profile included one severe and one mild ABCA4 variant, seen in 69% of patients. The most frequent allele c.5603A>T (p.Asn1868Ile) was found in 60.6%. Patient most often presented with decreased visual acuity (57.1%), with a median visual acuity of 0.8 Snellen decimals (IQR, 0.53–1.00) at first presentation. During follow-up 40.7% of the eyes showed foveal sparing, whereas 30.7% already had atrophy involving the fovea at initial presentation. Median time from symptom onset to foveal involvement was 15.4 years (95% CI, 11.1–19.6). The average decline in visual acuity was −0.03 Snellen decimals per year. Atrophy progressed at a median rate of 0.590 mm^2/year (IQR, 0.046–1.641 mm^2) for definitely decreased autofluorescence.
Conclusions: Late-onset STGD1 is associated with a milder genotype and phenotype, typically presenting with fundus flecks and foveal-sparing atrophy. This disease should be carefully differentiated from age-related macular degeneration, as misdiagnosis can lead to inappropriate treatment and prevent patients from accessing targeted therapeutic trials.

Restoration of central vision with a wireless subretinal photovoltaic implant in geographic atrophy due to AMD: Twelve-month results from the PRIMAvera trial

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Introduction: Geographic atrophy (GA) due to age-related macular degeneration (AMD) causes irreversible central vision loss once photoreceptors degenerate. No approved therapy restores lost central vision. The PRIMA system, a wireless 2 × 2 mm subretinal photovoltaic implant used with near-infrared projection glasses, aims to restore central visual function by stimulating inner retinal neurons within the atrophic area.
Patients and Methods: In this multicenter, prospective, baseline-controlled clinical trial, 38 patients with fovea-involving GA and baseline visual acuity ≥1.2 logMAR received a PRIMA implant. Best-corrected visual acuity (BCVA), with and without PRIMA glasses, was assessed at 6 and 12 months. Primary endpoints were (1) a clinically meaningful BCVA improvement of ≥0.2 logMAR at month 12 and (2) procedure- or device-related serious adverse events (SAEs).
Results: At month 12, 81% of evaluated participants (26/32) achieved a BCVA improvement ≥0.2 logMAR with the PRIMA system. Multiple imputation estimated an improvement rate of 80% across all implanted subjects. Mean BCVA gain with PRIMA glasses was 0.49 logMAR (~25 letters), with individual improvements up to 1.18 logMAR (59 letters). Central prosthetic vision was detected in 30/32 subjects. Twenty-six SAEs occurred in 19 patients, mostly transient postoperative events such as ocular hypertension and subretinal haemorrhage. No SAE was attributed to the implant alone. Natural peripheral visual acuity remained unchanged.
Conclusion: The PRIMA system provided meaningful central visual restoration in patients with advanced GA while preserving natural peripheral vision. These results support subretinal photovoltaic neurostimulation as a promising therapeutic approach for restoring central vision in atrophic AMD.

Involving patients in developing a patient-reported outcome measure for retinal detachment

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Purpose: To develop a retinal detachment-specific patient-reported outcome measure (PROM) that can measure quality-of-life impact, by involving patients.
Methods: Patients with a diagnosis of primary rhegmatogenous retinal detachment were recruited from the Rotterdam Eye Hospital (the Netherlands) and Newcastle Eye Centre (United Kingdom), using maximum variance sampling. Participants participated in a telephone interview using an interview topic list. Interview transcripts were coded into items (questions); items were grouped into quality-of-life (QoL) domains (themes) using an established ophthalmic QoL framework. Items similar to each other, too specific, or too unclear were removed. The final item list was compared with the literature.
Results: Seventy-three patients were interviewed, before or after surgery. We identified 1658 unique items. After 3 rounds of item refinement and cognitive interviews, the final item set included 355 items classified in 11 QoL domains: health concerns (45 items), emotional well-being including effects of emotional trauma (45), activity limitation (45), coping (39), visual symptoms (33), inconveniences (33), social well-being (28), ocular symptoms (23),

general symptoms (23), economic impact (22), and mobility (19). 288 items (81%) were new. There was 35% overlap with items developed for patients with macular hole, epiretinal membrane, and retinal vascular occlusions.
Conclusions: We have completed the first stage of developing a retinal detachment-specific PROM with patient involvement. Retinal detachment impacts heavily on patients' QoL. Most of the QoL impact is unique to retinal detachment and described for the first time.

Human amniotic membrane graft for closure of persistent macular holes with viscoelastic stabilization and air tamponade

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Purpose: To evaluate a series of patients with persistent macular holes (MH) treated with a mixed-gauge (25/23-gauge) pars plana vitrectomy after a prior vitrectomy with internal limiting membrane (ILM) peeling, in whom a non-closing MH persisted. The technique consisted of covering the MH with human amniotic membrane (hAM), combined with viscoelastic support and air tamponade.
Methods: Persistent MH was diagnosed using OCT. All patients had previously undergone vitrectomy with ILM peeling. During re-vitrectomy, a prepared hAM graft was placed over the macular hole and stabilized with a dispersive viscoelastic. An air tamponade was used. Postoperatively, patients maintained a reading position for one hour, followed by advice to avoid supine positioning during the first three days.
Results: Nine patients were included. In eight of the nine patients, the amniotic membrane graft remained in situ after the initial procedure. In one patient, graft displacement occurred, necessitating a reoperation, which resulted in successful macular hole closure. Postoperative OCT confirmed anatomical closure in all cases. Mean preoperative best-corrected visual acuity was 0.93 logMAR, improving to 0.56 logMAR at six weeks postoperatively, corresponding to a mean improvement of 0.37 logMAR. All patients demonstrated functional visual improvement. No intraoperative or postoperative complications were observed.
Conclusions: Using human amniotic membrane for persistent macular holes after prior vitrectomy, combined with viscoelastic stabilization, brief postoperative positioning, and air tamponade, appears to be a successful technique. This method may serve as an effective alternative to more invasive tamponade strategies for the management of persistent MH.

Outcomes of the sutureless scleral-fixated intraocular lens: A retrospective study

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Purpose: To evaluate intraoperative and postoperative outcomes of the Soleko FIL sutureless scleral-fixated intraocular lens (SSF-IOL) in patients with aphakia or insufficient capsular support.
Methods: Retrospective monocentric, study including 188 patients (191 eyes) who underwent SSF-IOL implantation. Intraoperative events, postoperative complications, refractive outcomes and intraocular pressure (IOP) were analysed.
Results: Primary indications were IOL (sub)luxation (44%) and aphakia after complicated phacoemulsification (31%). Mean surgery duration was 66.4 minutes (SD ± 23.6). Mean follow-up period was 433 days (range 15-2330). Intraoperatively complications occurred in 14 patients: four intraocular haemorrhages (2 iris, 2 sclerotomy-related) and haptic breaking in five cases. Additional isolated events included oversized sclerotomy, deep scleral flap, inadvertent anterior chamber entry, iridodialysis and one retinal tear during IOL positioning. At month 1, mean IOP was 15 mmHg (SD ± 4.9). Postoperative refraction was within ±1 diopter of target in 65.4% of eyes. Cystoid macular edema was the most common postoperative complication (17.3%) with 5/33 preoperatively present and 15/33 cases developing after the first month. Postoperative intraocular haemorrhage occured in 5.8% of which 5/11 related to transient hypotony with choroidal detachment and 10/11 spontaneous resolved. In 5 cases (2.6%) IOL tilted and three had a hypotony at first month (1.6%). In two patients (1%), a haptic extruded for which additional surgery was performed. Endophthalmitis or IOL opacification were not observed.
Conclusions: The Soleko FIL SSF-IOL provides a safe and effective option for eyes with aphakia or insufficient capsular support, with favourable refractive outcomes and a low rate of intra- and postoperative complications.

Reassessing the need for perfluorocarbon in retinal detachment surgery

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Purpose: Perfluorocarbon liquid (PFCL) is commonly used during retinal detachment surgery to facilitate subretinal fluid removal and protect the macula. However, its use raises environmental concerns. This study aimed to compare the clinical outcomes of retinal detachment surgery with and without the use of PFCL.

Methods: We conducted a retrospective cohort study of 1524 patients with primary retinal detachment who underwent 25-gauge pars plana vitrectomy at the Radboud University Medical Center in 2022–2023. Patients were stratified by macular involvement and proliferative vitreoretinopathy (PVR) grade. The primary outcome was the rate of recurrent retinal detachment, the secondary outcome was final post-operative visual acuity (VA).
Results: PFCL was utilized in 628 (41.2%) eyes and more frequently in macula-off detachments compared to macula-on (62.4% vs. 20.0%, $p<0.001$) and detachments with higher PVR grade (no PVR 18.1%, mild (grade A/B) 44.6%, severe (grade C/D) 63.3%, $p<0.001$). Within each PVR category, PFCL vs. no PFCL did not affect the rate of recurrent detachment: no PVR (12.1% vs. 6.0%, $p=0.083$), mild PVR (9.2% vs. 9.7%, $p=0.814$), and severe PVR (31.5% vs. 21.3%, $p=0.079$). Post-operative VA was worse in PFCL-treated eyes for both macula-on (median VA LogMAR 0.00 (Snellen 20/20) vs. 0.05 (Snellen 20/22), $p=0.028$) and macula-off detachments (median VA LogMAR 0.30 (Snellen 20/40) vs. 0.40 (Snellen 20/50), $p=0.015$), however, this was not significant after adjusting for PVR grade.
Conclusions: In this study PFCL did not reduce recurrent detachment rate or improve post-operative VA. Surgical success in retinal detachments can be achieved without the use of PFCL.

Visualization of photoreceptors after vitrectomy for retinal detachment using adaptive optics

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Purpose: Adaptive optics (AO) imaging is an expanding tool in retinal assessment. This ongoing pilot study aims to investigate its feasibility in patients who have undergone vitrectomy for retinal detachment.
Methods: A total of 20 patients who underwent (phaco) vitrectomy for macula-off or split fovea rhegmatogenous retinal detachment have been enrolled. AO imaging was performed using an adaptive optics flood-illumination ophthalmoscopy (AOFIO) system (rtx1 retinal camera, Imagine Eyes, France). The primary outcome measures are visualization of photoreceptors and identification of potential imaging biomarkers at 7 weeks and 6 months postoperatively.
Results: To date, 18 patients have undergone AO imaging, and six patients have reached the 6-month follow-up. In 16 patients, photoreceptors were clearly visible in some retinal areas 7 weeks postoperatively, whereas other regions appeared blurred. In two patients, adequate imaging was not possible due to the presence of a trifocal intraocular lens and high axial length (29 mm). One patient could not be imaged 6 months postoperatively with AOFIO because of cataract (grade 3). In five patients with analyzable 6-month images, larger areas with visible photoreceptors and higher cone density values were observed compared with the 7-week visit. These changes may represent a potential progression biomarker.
Conclusions: Photoreceptor imaging with AOFIO is feasible at both 7 weeks and 6 months after vitrectomy. Optimization of the imaging protocol, refinement of case selection, and improved interpretation of postoperative photoreceptor changes will be important for future studies and clinical integration.

Spike in intraocular pressure a few weeks after goniotomy for uveitis: No cause for apprehension

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Purpose: To study the intraocular pressure (IOP) after a goniotomy procedure in patients with uveitis during the first 3 months after surgery.
Methods: A multicenter retrospective analysis was performed. We included 36 eyes of 26 uveitis patients who underwent goniotomy due to uveitis-related oculair hypertension or glaucoma between 2011 and 2023.
Result: Median IOP before goniotomy was 32 (interquartile range 26–37) mmHg and the median IOP 1 day after goniotomy 12 (9-14) mmHg. Between postoperative weeks 1 and 6, 27 of 36 eyes experienced an increase in IOP above 20 mmHg, with the peak IOP recorded at 30 (26–35) mmHg. This was mainly controlled by topical glaucoma medication, but led to a repeated goniotomy in 3 cases (due to a continuing elevated IOP beyond 6 weeks after goniotomy). We observed these 3 failures, only among patients aged 22 or older (3 of 7 patients). In the remaining 33 eyes, the median IOP 3 months after goniotomy was 17 (15–18) mmHg with a median of 2 different types of topical glaucoma medication (0–3). There was no severe recurrent uveitis observed in this period.
Conclusions: Goniotomy in uveitis patients is a safe and effective procedure. There were no failures in patients aged 21 and under. A spike in IOP between week 1 and 6 was observed in the majority of eyes, which was nearly always manageable with topical glaucoma medication, and is thereby no sign of failure.

*HLA-DRB1*15:01* paediatric uveitis and the risk of MS

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Purpose: The purpose of this study is to characterize the clinical presentation of the *HLA-DRB1*15:01* positive non-anterior uveitis phenotype in children and to determine if early markers of MS are present in this subgroup.
Methods: In a cohort of 65 patients with non-infectious non-anterior paediatric uveitis patients we retrospectively characterized the uveitis phenotype based on underlying *HLA-DRB1*15:01* genotype and described the clinical parameters on OCT and fluorescein angiography (FA).
Results: Patients with *HLA-DRB*15:01*-positive uveitis had almost three times more intermediate uveitis compared to patients with *HLA-DRB*15:01*-negative uveitis (70.0% versus 24.4% respectively, $p=0.002$). Vitreous floaters were found more frequently in *HLA DRB1*15:01*-positive compared to -negative cases (80.0% versus 53.3% respectively, $p=0.041$). Less cells in the anterior chamber were seen in *HLA-DRB*15:01* positive patients compared to in *HLA-DRB*15:01* negative patients (55.0% versus 90.7% respectively, $p=0.001$). Fern-like capillary leakage was almost twice as common in *HLA-DRB*15:01* positive compared to *DRB*15:01* negative patients (64.7% versus 34.1% respectively, $p=0.032$). Multiple sclerosis developed in one *HLA-DRB*15:01* positive patient (5.0%), 15 years after uveitis was diagnosed.
Conclusion: The *HLA-DRB1*15:01* allele is associated with idiopathic intermediate uveitis in children with absence of anterior chamber cells, presence of vitreous floaters and peripheral capillary fern-like leakage as distinct clinical features. The usefulness of *HLA-DRB1*15:01* typing for predicting the risk of multiple sclerosis is yet to be determined.

Comparison of conventional and biological DMARDs in non-infectious scleritis

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Purpose: To compare the efficacy of conventional synthetic disease-modifying antirheumatic drugs (csDMARDs) versus biological DMARDs (bDMARDs) in therapy-naïve patients with non-infectious scleritis, and to identify possible clinical determinants influencing treatment response.
Patients and Methods: In this retrospective cohort study, we reviewed data from therapy-naïve patients with non-infectious scleritis, subsequently treated with csDMARDs or bDMARDs in the Erasmus University Medical Centre, Rotterdam, between February 2002 and July 2025. Baseline and clinical characteristics including treatment response and time to quiescence (inactive disease on ≤7.5 mg prednisolone/day) were extracted from medical records.
Results: Forty-four therapy-naïve patients with non-infectious scleritis treated with csDMARDs ($N=30$) or bDMARDs ($N=14$) were included. Quiescence of scleritis was achieved in 20/30 (67%) of patients treated with csDMARDs versus 11/14 (79%; bDMARDs; $p=0.498$), with similar median time to quiescence (114.5 vs 109.0 days; $p=0.650$). Relapse rate of scleritis activity was 5/20 (25%) in patients treated with csDMARDs versus 3/11 (27%; bDMARDs; $p=1.000$) with comparable duration between quiescence and relapse (150.0 vs 145.0 days; $p=0.881$). Start of treatment within 60 days after diagnosis yielded faster quiescence (median 93.5 vs 170.0 days, $p=0.045$). No clinical determinants were found to be associated with treatment response, other than that a higher BMI correlated with a longer time to quiescence in patients treated with bDMARD ($p=0.032$).
Conclusions: Our results show no significant difference in treatment response in non-infectious scleritis between csDMARDs and bDMARDs. Treatment initiation within 60 days after diagnosis shortens time to quiescence of scleritis, supporting fast intervention.

Efficacy and safety of vamikibart in patients with uveitic macular edema: First report of phase 3 MEERKAT/SANDCAT trials

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Purpose: Interleukin-6 (IL-6) is a key inflammatory mediator in the pathogenesis of uveitic macular edema (UME), a vision-threatening complication of noninfectious uveitis (NIU). The objective is to evaluate the efficacy and safety of Vamikibart, a recombinant, humanized, intravitreal (IVT), monoclonal anti–IL-6 antibody, in NIU-UME.

Methods: MEERKAT (MK; NCT05642312; $N=245$) and SANDCAT (SC; NCT05642325; $N=256$) are identical global phase 3 randomized trials. Patients with NIU-UME were allocated to IVT vamikibart 0.25 mg, 1.0 mg, or sham (1:1:1); 4× monthly IVT were given, then PRN to W48. Primary endpoint at W16 was the proportion of patients with ≥15 letters BCVA gain from baseline.
Results: Vamikibart showed clinically meaningful efficacy vs sham at W16. Proportion of patients with ≥15-letter improvement in BCVA for MK/SC: 26.1/34.0% 0.25 mg, 43.2/24.2% 1.0 mg vamikibart vs 6.3/13.3% sham. Mean change from baseline (CFB) BCVA for MK/SC: +9.6/11.9 letters 0.25 mg, +12.8/9.2 letters 1.0 mg vamikibart vs +3.5/5.0 letters sham. Mean CFB CST for MK/SC: –187.5/–209.7 µm 0.25 mg, –196.1/–194.7 µm 1.0 mg vamikibart vs –58.5/–43.5 µm sham. Vamikibart was well tolerated; serious ocular AE incidence was low.
Conclusions: Vamikibart, a novel IVT anti–IL-6 therapy, was efficacious and well tolerated in patients with UME.

Comparison of serological and molecular diagnostic analysis of vitreous and aqueous humor of patients with vitreoretinal lymphoma

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Purpose: To evaluate the diagnostic and prognostic value of combined serological and molecular analysis of vitreous and aqueous humor of patients with (primary) vitreoretinal lymphoma ((P)VRL).
Methods: In this retrospective cohort study, we reviewed data from patients with suspected (P)VRL who underwent vitreous biopsy and/or anterior chamber fluid sampling at the Erasmus University Medical Centre in Rotterdam between April 2015 and August 2025. Clinical characteristics and diagnostic results, including flow cytometric immunophenotyping, cytomorphology, myeloid differentiation factor 88 (MYD88), IL-10/IL-6 cytokine ratio were extracted from medical records. Missing laboratory results were supplemented by analyzing residual stored material, when available.
Results: A total of 44 patients with (P)VRL (biopsy-proven by cytomorphology and/or flow-cytometric immunophenotyping), comprising 34 vitreous biopsies and 29 anterior chamber fluids, and 145 control patients, with 145 vitreous biopsies and 24 anterior chamber fluids, were included. MYD88, in either anterior chamber fluid or vitreous, showed a sensitivity of 67% (26/39) for (P)VRL and a specificity of 99% (103/104). IL-10/IL-6 cytokine ratio showed a sensitivity of 94% (50/53) for (P)VRL and a specificity of 100% (144/144).
Conclusions: Our results demonstrate high sensitivity and specificity of the IL-10/IL-6 cytokine ratio, and high specificity of MYD88, for the diagnosis of (P)VRL. Combining these diagnostic markers can further strengthen clinical suspicion. Additional analyses are needed to evaluate the diagnostic performance of these markers in vitreous versus anterior chamber fluid.

Sutureless Müller's muscle-conjunctival resection for the correction of ptosis: A case series

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Purpose: To report the results of sutureless Müller's Muscle-conjunctival Resection (sMMCR) in terms of efficacy, lower eyelid position, safety and adverse effects. The Sutureless Müller's Muscle-conjunctival Resection (sMMCR) is a posterior-approach ptosis correction technique that utilizes a graded resection of the Müller's muscle and conjunctiva without the use of permanent or absorbable sutures. In this modified procedure, anatomical adherence of the excised tissue margins is achieved through electrocoaptation, which employs thermal energy to fuse the wound edges, thereby reducing postoperative foreign body sensation and eliminating suture-related complications.
Methods: Retrospective case series. Surgical outcomes including margin reflex distance (MRD) 1 improvement, change in lower eyelid position, symmetry and complications, were analyzed preoperatively and three months postoperatively. Linear mixed models were used to account for inter-eye correlation.
Results: In total, 100 eyelids of 61 patients were included, of which 45 patients (73.8%) had involutional ptosis and 16 (26.2%) had contact lens-induced ptosis.
Mean ΔMRD1 and ΔMRD2 at 3 months were 2.35 ± 1.13 mm and -0.45 ± 1.00 mm, respectively. Symmetry within 1 mm for bilateral eyes was observed in 86.3% of cases. Without concurrent procedures, sutureless MMCR took 2 to 3 min to perform. No perioperative or postoperative complications occurred.
Conclusions: The sutureless MMCR is an effective and safe treatment for mild to moderate ptosis in patients with adequate levator function (>10 mm). Its high efficacy and significantly reduced operative duration suggest this procedure has the potential to become the standard of care for ptosis correction.

Sutureless mullerectomy in the aesthetic oculoplastic practice

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Purpose: The purpose of this study is to describe the advantages and disadvantages of a sutureless technique for correcting mild to moderate eyelid ptosis through futureless Conjunctiva Müller Muscle Resection (CMMR).
Methods: We describe a series of 25 eyes of 16 patients operated on by a single senior surgeon in various private

practices in Spain and the Netherlands. The principles of CMMR were followed in a standardised manner, using the 4:1 rule; 4 mm of resection for every 1 mm of desired correction. The difference between the classic technique and this approach is that, once the Putterman clamp is placed on the conjunctiva and Müller′s muscle, the tissue is removed directly without previous suture placement. We also do not place a haemostat clamp before excising. Hemostasis of the wound is then performed, and the tissues in the upper fornix are repositioned. We documented at baseline and at 8 weeks postoperatively: distance from the upper eyelid margin to the pupillary light reflex (MRD1 in mm), the palpebral fissure height (PF in mm), and standardised portret photographs.
Results: We performed the surgery in a total of 25 eyes and achieved a mean increase in MRD1 of 1.24 mm (range 0 to 3 mm) and a mean increase in PF was 1.30 mm (range 0 to 4.5 mm). In one eye of one patient, no significant difference in MRD1 was observed after surgery. In addition, one other patient developed a pyogenic granuloma in one eyelid and exhibited a slight irregularity in the eyelid contour postoperatively.
Conclusions: Ptosis correction using a sutureless CMMR is an effective way of lifting the eyelid margin, prevents potential corneal damage caused by sutures, and shortens the surgical time. In our opinion it's an ideal adjunct procedure during upper eyelid blepharoplasty.

Seronegative euthyroid Graves' orbitopathy does not exist

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Purpose: Graves' orbitopathy (GO) typically occurs in hyperthyroid patients with detectable TSH-receptor antibodies (TSH-R-Ab). A small subset develops GO while euthyroid (EGO). Even more rare is the subgroup of EGO patients without detectable TSH-R-Ab (seronegative). However, seronegativity in these cases is solely determined by a negative thyrotropin-binding inhibitory immunoglobulin (TBII) assay. Therefore, in this study we explore the use of a novel and highly sensitive bioassay (Turbo TSI) to detect TSH-R-Ab in TBII-negative EGO. Furthermore, we describe a large cohort of TBII-negative EGO patients and compare clinical characteristics to hyperthyroid GO patients.
Methods: We retrospectively reviewed 20 patients with TBII-negative EGO (EliA TBII). If available, biobank serum was used to reassess TSH-R-Ab levels with Turbo TSI bioassay and with a bridge-based TSI binding immunoassay. Clinical features of TBII-negative EGO patients were compared to 40 hyperthyroid GO controls.
Results: All 20 patients were euthyroid and TBII-negative at presentation. Biobank serum was available for six individuals; two were positive with the bridge-based assay, whereas all six tested positive with the Turbo TSI bioassay. Compared with hyperthyroid GO, TBII-negative EGO patients exhibited more unilateral or asymmetric involvement, milder and less active disease, and required fewer surgical interventions.
Conclusions: Apparent seronegativity in TBII-negative EGO reflects limitations of standard binding assays rather than true absence of TSH-R-Ab. The Turbo TSI bioassay offers higher sensitivity with a simple and rapid protocol, improving diagnosis in these patients. TBII-negative EGO presents with a milder phenotype than hyperthyroid GO.

Frontalis flap: A material-free alternative to the frontalis suspension

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Background: The frontalis suspension (autologous fascia lata or silicone) is the standard of care in the Netherlands and many other countries for severe ptosis with poor levator function, but it has material-related complications and sometimes suboptimal dynamics. The frontalis flap—a local advancement of the frontalis muscle without an implant—offers a potential alternative.
Objective: To discuss therapeutic options for the treatment of blepharoptosis with poor levator function, including a technique first described in 1901 but recently (in 1982) modified: the frontalis flap. This technique is popular in Asia and is becoming more popular in Europe and the US.
Methods: Review of various surgical techniques that can be applied to blepharoptosis with poor levator function. The indications and practical implementation of the frontalis flap will be discussed using step-by-step photographs. Initial experiences with this technique in patients at the Rotterdam Eye Hospital since 2024 will be discussed.
Results: Existing techniques for treating blepharoptosis with poor levator function are frontalis suspension with autologous (own fascia lata) or heterologous material (donor fascia lata, silicone rubber, sutures, Mersilene mesh) and the frontalis flap. Immediate postoperative management focuses on corneal protection. Drops/ointment and taping or Frost sutures can be used. Initial

experiences show promising results, but lagophthalmos, postoperative hematoma formation, and undercorrection are possible.

Conclusion: Frontal suspension remains a good method for treating poor levator function, but the frontalis flap is an interesting material-free alternative.

Augmented reality–guided orbital decompression: A pilot study using phantom heads

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Background: Thyroid Eye Disease (TED) often requires orbital decompression surgery, a technically challenging procedure where conventional navigation is rarely used due to ergonomic limitations. Head-Mounted Display Augmented Reality (HMD-AR) is a novel technique that projects 3D images directly into the surgical field, potentially improving spatial orientation without requiring the surgeon to shift their gaze. This pilot study evaluated the workload, usability, and surgical accuracy of HMD-AR compared to conventional CT-based planning.

Methods: One experienced orbital surgeon and two medical students performed decompressions on phantom skulls. The study compared AR-guidance against standard CT-guidance. Workload was assessed using the NASA-TLX questionnaire and usability via the USE questionnaire. Surgical accuracy was quantified using the Jaccard Index (JI) to measure the overlap between planned and executed decompression surfaces, while procedural time was also recorded.

Results: Participants reported a lower overall workload using the AR method compared to the conventional approach. After correcting for technical failures, procedural time decreased by 37% with AR. Regarding accuracy, there was no statistically significant difference between AR (35.2%) and CT guidance (37.5%). However, usability scores improved over time, and the experienced surgeon demonstrated a marked learning effect within the AR condition, improving accuracy from 16.3% to 51.4%.

Conclusions: Implementing HMD-AR leads to reduced workload and shorter operative times. While initial accuracy is comparable to standard methods, the observed learning effect suggests that AR can enhance procedural precision and spatial orientation once familiarity is achieved. These findings support further research into AR as a navigational tool in orbital surgery.

Timely corticosteroid treatment can improve functional craniofacial fibrous dysplasia associated cranial nerve involvement – A meta-analysis

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Purpose: In craniofacial fibrous dysplasia (CFD), expansion of soft bone tissue can cause sudden onset cranial nerve entrapment. This might be improved by glucocorticoids. Therefore we aimed to explore the potential benefits of using corticosteroids in the case of CFD associated cranial nerve impairment.

Methods: From a large cohort of patients with CFD we identified patients with cranial nerve impairment who received corticosteroid therapy and collected data on clinical features, treatment characteristics, and its effect. In addition, we conducted a comprehensive literature search and identified the reports of cases with CFD associated cranial nerve impairment who received corticosteroid therapy.

Results: From our cohort of 185 CFD patients we identified four patients with cranial nerve impairment resulting from CFD who received treatment with corticosteroids. In two cases the abducens nerve, in one case the optic nerve and in one case the trigeminal nerve was involved. Corticosteroid treatment did not relieve trigeminal nerve pain. In two cases of abducens involvement, the treatment resolved diplopia, although in one case, there was no more effect after the third recurrence. In the case of optic nerve involvement, visual function improved. This is in line with available literature, in which six out of nine cases identified showed a beneficial effect of corticosteroid treatment.

Conclusion: Available data on the effect of corticosteroids on cranial nerve impairment associated with CFD is very limited but most reported cases show a beneficial effect in these pre-selected group. However, the effect appears to be more pronounced in functional neuropathies rather than painful neuropathies.

Insight in radiation dosing in orbital MALT lymphoma patients

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Introduction: Mucosa-associated-lymphoid-tissue (MALT) lymphoma of the orbit is often treated with radiotherapy after hematological staging. Recommended radiation dosages are 24–36 Gy for curation in localized stages, and doses as low as 4 Gy as a palliative option in disseminated disease. Recent studies however suggest lower dosing for mild stages, with reasonable cure rate and fewer side effects. With this pilot study we want to explore the radiation dosing strategies for patients with orbital-MALT-lymphoma at the Amsterdam UMC.

Patients and methods: Retrospective case series of patients with orbital-MALT-lymphoma, diagnosed by biopsy, at the Amsterdam Orbital Center. Staging of the disease and treatment regimens were extracted from the electronic database.

Results: Twenty-two cases were classified as orbital MALT-lymphoma. Thirteen patients showed orbital disease only (stage I), 2 local stage II, 5 disseminated (stage IV), and 2 unknown stage. Higher radiation dose (12 × 2 Gy) was given in 7 cases (all stage I). Low dose (2 × 2 or 1 × 4 Gy) was given in 11 cases (4× stage I, 2× stage II, 5× stage IV, 4 not available). All 7 cases with 12 × 2 Gy-regimen had good initial response, one had a recurrence. Two out of 11 cases with low-dose radiotherapy received additional radiotherapy due to insufficient effect. Cataract and dry eyes are more often documented for the 12–2 Gy-group.

Conclusion: This pilot study supports the data in the literature that there might be an place for lower dosing in local orbital lymphoma cases. Prospective trials including ophthalmologist, hematologist and radiotherapist will give more detailed insight.

Circulating tumor cells in uveal melanoma: Multi-marker detection and association with disease state

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Purpose: Uveal Melanoma (UM) is primarily treated with eye-sparing radiotherapy, leaving limited tumor tissue for molecular analysis. Circulating tumor cells (CTCs) may offer a minimally invasive alternative for genomic tumor profiling. This pilot study evaluated the feasibility of a multi-marker CTC capture approach in UM patients at diagnosis, during fractionated stereotactic radiotherapy (fSRT), and at metastatic progression.

Methods: Patients with localized or metastatic UM were prospectively enrolled. Peripheral blood samples were collected at baseline, during fSRT, and upon detection of metastases. CTCs were captured and enumerated using a multi-marker approach and fluorescence microscopy.

Results: A total of 76 patients were included: 68 with localized and eight with metastatic disease. Four initially presented with localized disease but developed metastasis during follow-up. For comparisons, only their metastatic samples were used, yielding 64 localized and 12 metastatic samples. CTCs were detected in 69.1% of patients with localized disease at baseline and in 83.3% of those with metastases. CTC counts were significantly higher in metastatic than localized disease (median 8 vs. 3; $p = 0.010$). Among patients undergoing fSRT, paired analysis showed a significant increase in CTC counts between day 3 and 5 (median 1 vs. 4; $p = 0.007$). No significant associations were observed between baseline CTC counts and tumor thickness, largest basal diameter, tumor volume, AJCC tumor stage, tumor location, or molecular risk class.

Conclusions: Multi-marker CTC detection in UM patients is feasible across disease stages. Increased CTC counts during fSRT may offer a window for molecular characterization. Larger studies with longitudinal blood sampling are needed to validate clinical and prognostic utility.

Radiation therapy for invasive conjunctival squamous cell carcinoma

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Purpose: To describe a series of 3 patients with squamous cell carcinoma (PCC) of the inferior fornix invading the anterior orbit.

Methods: Since September 2024 three patients were referred to the Leiden University Medical Center with an local invasive PCC of the inferior fornix. One patient had previously undergone surgical excisions and topical chemotherapy. In two cases the tumor extended into the anterior orbit. Standard curative treatment would involve surgical resection with or without adjuvant radiotherapy. However, surgery carries significant risks due to the proximity of adjacent structures. As an alternative, all patients were treated with curative radiation therapy 70 Gy in 35 Fractions with VMAT Photon therapy.

Results: MRI demonstrated a good response in all patients, 8 to 10 months after irradiation therapy. Short term ocular side effects were mild. No patient developed distant metastasis.

Conclusions: In squamous cell carcinoma invading the anterior orbit, radiation therapy can be an effective alternative treatment for extensive surgery.

Acta Ophthalmologica

Conditions for transfer of knowledge of and interest in the history of ophthalmology, regarding means of communications and activities

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Purpose: Transfer of ophthalmic historical knowledge to future ophthalmologists is important because they maintain a connection with the past, understand how ophthalmology evolved, and can identify patterns and trends to better prepare for future challenges and opportunities. Ophthalmic historical societies are facing decline of membership and attendance at annual meetings. The question arises which weak areas makes them vulnerable.

Methods: A systematic inventory was made of barriers, conditions and facilitators. We distilled the principles that underly educational study and teaching of history, and gathered items applicable to historical societies. These were grouped into two domains: communications and activities, and worked out for application to ophthalmology.

Results: We formulated essential means of communication: Website with documents, images, conference reports, videos, and podcasts. Ophthalmic historical publications on website. Catalogs of ophthalmic instruments on website. The story behind ophthalmological facts: the human side, personal stories, intrigues and conflicts.

We formulated essential activities: Ophthalmic historical meetings or sessions at general ophthalmological congresses, lectures, exhibitions, book presentations, and workshops. Visits to ophthalmic historical sites. Collaboration between ophthalmologists, ophthalmic physicists and other scientists. Working out links between current issues and ophthalmic historical events

Conclusions: The transfer of knowledge of and interest in the history of ophthalmology to young ophthalmologists and ophthalmic physicists is impeded most by the lack of website communication, due to the members' age, their decreased income, lack of earnings in history and lack of support by companies. This should be resolved by website hosting, offered by national ophthalmological societies.